Design and Analysis of a Single Axis Capacitive Accelerometer Sensor for Range of Motion Applications

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***Abstract*—This report presents the development and analysis of a single-axis capacitive accelerometer, designed for ROM monitoring in rehabilitation. Based on the MEMS technology, this sensor operates with differential capacitance created by displacing a proof mass, hence changing the capacitances between the electrodes. This module is designed to measure accelerations from -4g to +4g in a minimum bandwidth of 5 kHz with high sensitivity, including the possibility to adapt to low-frequency motion tracking. A structured methodology has been followed: from finite element simulations in ANSYS Workbench, the mechanical and electrostatic behaviors of the sensor are modeled. Stable performance, free from resonance, has been confirmed by both modal and linear perturbation analyses, meeting the key requisite for capturing slow and controlled movements typical in rehabilitation. The design order has been reduced using ROM, thus allowing dynamic simulations and system-level integration with Twin Builder. The capacitance changes are converted into voltage by an analog circuit designed in both Twin Builder and LTSpice. Moreover, the Transistor-level logic is designed in Cadence Virtuoso to realize the physical gain of the system at 120dB with source degeneration and a Two-Stage Amplifier. The voltage output of the analog circuitry in terms of frequency is sensed by a voltage-controlled oscillator (VCO) which is further fed into frequency measurement and decision circuit in the digital domain. The smart integration enables a device that is capable of measuring real-time data-driven monitoring and feedback of remote and personalized rehabilitation therapies.**

***Index Terms*—FEM, Linear-perturbation analysis, Cadence virtuoso, Reduced Order Modeling, Digital Twin, Therapy Monitoring**

I. Introduction

Accelerometer, as a MEMS device is a critical sensor in consumer products like smartphones, activity trackers and automotive applications. This device can be used for activity monitoring; specifically range of motion measurement for rehabilitation purposes. The accelerometer can be used to measure the degree of motion with respect to displacement.

Progressively increasing the range of motion (ROM) during rehabilitation is crucial to prevent muscle and joint stiffness, to maintain or restore flexibility, and to prevent the formation of scar tissue. Rehabilitation can be challenging due to human error, inconsistency, and limited real-time data. Smart sensors could offer access to accurate and objective ROM measurements. They could track progress consistently, identify deviations, and alert therapists to potential issues and open the chance for remote monitoring, progress tracking and providing real-time, data-driven insights to guide and personalize the rehabilitation process.

Previously, a surface-machined accelerometer for the measurement of the range of motion was designed, as the sensing mechanism for our rehabilitation device. The required measurement range is from -4g to +4g, which should have a minimum bandwidth of 4 kHz for the general operation range. The accelerometer is of differential capacitance type and it is chosen based on the benefits in sensitivity, temperature stability and design flexibility it offers.

Currently, the objective of this project is to develop a comprehensive system-level design that integrates these principles into a functional sensor for ROM measurement. This process begins with ANSYS simulations to model the sensor's physical and electrical behavior, progresses to system-level modeling using Twin Builder for dynamic and circuit-level simulations, and culminates in the implementation of both analog and digital system designs to ensure robust functionality and adaptability.

II. Background

The accelerometer sensor operates on the principle of differential capacitance, where the displacement of a proof mass suspended by folded beam springs alters the capacitance between electrodes. Each finger of the proof mass, positioned between two electrodes, acts as a common plate, forming two capacitors in series, as illustrated in Figure 1. When the movable finger is at centre, the distance between movable finger and electrodes on either side is same (d). The gap between the movable finger and electrodes on either side upon the displacement x is d_1 and d_2. The gap d_1 is equal to d + x and d_2 is equal to d - x. The displacement of the finger creates a capacitance difference between the two capacitors, which is proportional to the acceleration. This relationship of a parallel plate capacitor is described by Eqn. 1, where C represents the capacitance for a given distance d, overlap area A, and permittivity ϵ.

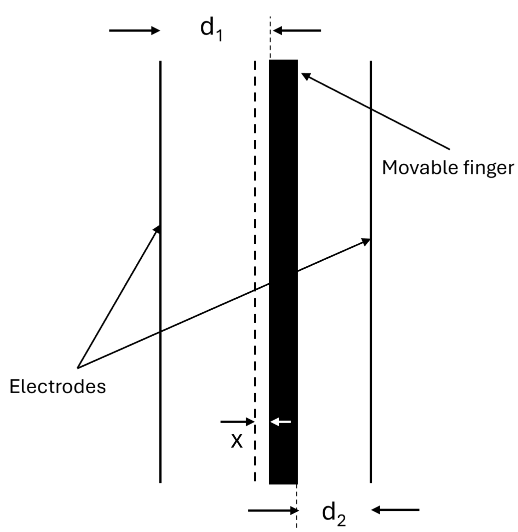

Fig. 1. Illustration of electrodes and finger as differential capacitors

$$C = \frac{\epsilon A}{d} \tag{1}$$

$$C1 = \frac{\epsilon A}{d + x} \tag{2}$$

$$C2 = \frac{\epsilon A}{d - x} \tag{3}$$

$$\Delta C = \frac{2\epsilon A x}{d^2 - x^2} \tag{4}$$

$$\Delta C = \frac{2\epsilon A x}{d^2} \tag{5}$$

The individual capacitances $C1$ and $C2$, corresponding to the left and right sides of the electrodes, are given by Eqns. 2 and 3, respectively, Where x is the variable displacement. The differential capacitance ΔC, shown in Eq. 4, is non-linear. However, when $x \ll d$, the approximation in Eq. 5 holds, providing a linear relationship between the displacement and the capacitance difference. The inertial force acting on the system is defined in Eq. 6, where m is the mass of the proof mass, and a is the acceleration. According to Hooke's law, the relationship between the force applied by the spring and the displacement is expressed in Eq. 7, where k is the spring constant. Combining these, the deflection of the proof mass x can be expressed in terms of the acceleration, as shown in Eq. 8.

$$F_{inertial} = ma \tag{6}$$

$$F_{damping} = kx \tag{7}$$

$$x = \frac{ma}{k} \tag{8}$$

By substituting Eq. 8 into the linearized differential capacitance equation (Eq. 5), a direct relationship between the differential capacitance ΔC and acceleration a can be derived. This relationship shows the differential capacitance as proportional to the applied acceleration.

$$\Delta C = \frac{2\epsilon A m a}{k d^2} \tag{9}$$

The accelerometer design incorporates a folded beam spring structure, where folded beams are connected in series to form a guided spring system. The total spring constant (k) for one side of the beam structure is calculated using Eqn. 10, while the total spring constant of the sensor (k_T) is derived in Eqn. 11.

$$k = \frac{(\pi^4/6) E w h^3}{(2L_1)^3 + (2L_2)^3} \tag{10}$$

$$k_T = 2k \tag{11}$$

III. Reduced order modelling of the MEMS device

This section deals with the design of MEMS accelerometer and the approximation of dynamic modeling to realize the application of range of monitoring for rehabilitation. The following sections highlight the iterative design modeling from finite element approach to reduced order modeling.

A. Simulation outline

- FEM Coupled Simulation: This simulation in ANSYS is conducted to numerically solve for the deflection of the capacitive accelerometer model under the influence of acceleration load. The operation of the device is decided through applied bias voltage of (-1.5V and +1.5V). This helps to numerically solve the geometry issues and performance of the device under test. The meshing is done through element sizing, edge sizing and so on. The model is linearized around the equilibrium points where electrostatic forces are equal to mechanical forces to reduce the complexity of the design. The FEM simulation is aimed to produce the electromechanical coupled response and behavioral response of the accelerometer, which can be used to validate the functionality, before fabricating the device.

- Electrostatic Analysis: In total, three cases are devised to see the performance. *Case-I* deals with a pure mechanical response of the system without any electric load. *Case-II* introduces the electrostatic forces only. This case is important because in the our model, the deflection at this stage should ideally be zero since electric voltage (as load) does not contribute in deflection. The *Case-III* realizes the coupled simulation. where the effect of both electrical and mechanical domain is coupled through 8-node and 20-node elements (Solid186, Solid226). The type of coupling used is direct coupling and relation of electromechanical domain is sub-categorized as; the acceleration load as input produces deflection, and deflection changes the capacitive of the comb-fingers. This capacitance is sensed through Analog and Digital circuitry designed at later stages of this report.
- Modal analysis: The required frequency of our application is considered to be of maximum range 500 Hz -1kHz [1] that successfully monitors the slow motion activity which is essential for rehabilitation. The resonant frequency of the accelerometer used in this study is 4 kHz, which shows its peak sensitivity to vibrations of that particular frequency. This is quite important because it will ensure that the data to be acquired will be very accurate, while the normal range of motion in rehabilitation settings is made up of low-frequency movements. A high resonant frequency ensures that, during these low-frequency activities, the accelerometer is operating well outside its resonance zone, therefore reducing chances of signal distortion. This makes the device highly suitable for monitoring range of motion in rehabilitation, since it can capture with precision the accelerations and movements without the interference of resonance effects. The accelerometer was also kept away from mechanical vibrations at or near its resonant frequency when testing in order not to compromise the accuracy of the data. In summary, the high resonant frequency supports robust performance in applications requiring low-frequency motion analysis.
- Linear Perturbation Analysis: In a physical world, the system deals with non-linearity. This is done through mode superposition to mimic the behavior and solve numerical values through perturbation based on the applied load. When biased voltage is applied, it changes the system response in terms of frequency (i.e. reduction of frequencies on a biased voltage).
- ROM Generation: ROM is generated using post processing APDL commands. For reduced order modeling the input of the model is acceleration and output is deflection. The design of the device is symmetric and we choose 5 output (deflection) points to verify the behavior of the device. These five points are chosen are decided on a location of center of mass plate, right edge of mass plate, left edge of mass plate, one of comb fingers edge, and middle of comb finger.
- ROM Analysis in Twin Builder: ANSYS Twin Builder is utilized to add generated ROM model as mechanical component. The model is actuated through the applied load of acceleration ranging values from (-4, 4 g) and deflection of the device is noted. The vibrational response of the ROM model is analysed and the frequency is compared with the natural frequency from modal analysis.

B. Material properties

The materials used in the simulation are polysilicon and air, as illustrated in Figure 2. Their respective properties are summarized in Table I. The air is modelled as "elastic air" with zero poisson ratio and small Young's modulus as recommended in [2]. The Young's modulus of air is calculated using the formula in Eq. 12, which is from [2].

$$E_{\text{air}} = \left(\frac{U_{\max}}{d_{\min}}\right)^2 \cdot \left(\frac{\epsilon_0}{200}\right) \tag{12}$$

where $U_{\max}$ is the maximum voltage (1.5 V), $d_{\min}$ is the minimum gap distance (1 μm), and ϵ_0 is the permittivity of free space.

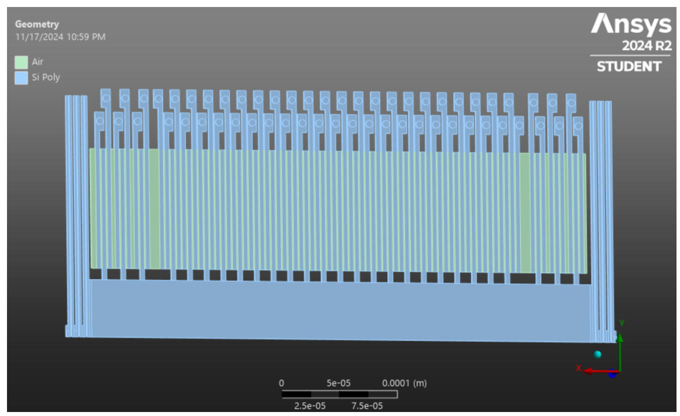

Fig. 2. Materials used in The Simulated Sensor

TABLE I
MATERIAL PROPERTIES OF POLYSILICON AND AIR [2] [3]

Property	**Polysilicon**	**Air**
Density (kg m^{-3})	2330	1.23
Young's Modulus (GPa)	160	9.96E-11
Poisson's Ratio	0.22	0
Relative Permittivity	–	1

C. Geometry

The geometry of the sensor was originally designed as shown in Figure 3, as it includes a proof mass, functioning as the suspended mass sensitive to motion, 54 sensing fingers, connected to the proof mass to enable differential capacitance and folded beam springs. The folded beam structure has a length of 420 μm and the width of 2 μm, which, together with the spring beam, supports displacement and optimal resonance frequency. The capacitance gap of 1 μm between electrodes is designed to ensure sensitivity to the proof mass's displacement. Previously, 150 perforations were incorporated into the proof mass, each with a 4 μm square hole, allowing airflow through the structure, However, in this ANSYS simulation, they were not modeled, and the air domain between the solid structures was considered and added to the geometry, with dimensions of 104 μm by 445 μm.

Additionally, due to the symmetric structure of the accelerometer, only half of the geometry is modeled to reduce computational load. In ANSYS, symmetry boundary conditions are applied to simulate the full response accurately while using just the essential portion of the design. This approach maintains the structural integrity of the simulation results while optimizing resources. Figure 4 illustrates the overall geometry of the sensor, indicating all parts of the design. Figures 5 and 6 provide a clearer view of the springs, anchors and sensing fingers structures. Furthermore, Table II provides a summary of the dimensions and key parameters of the accelerometer design.

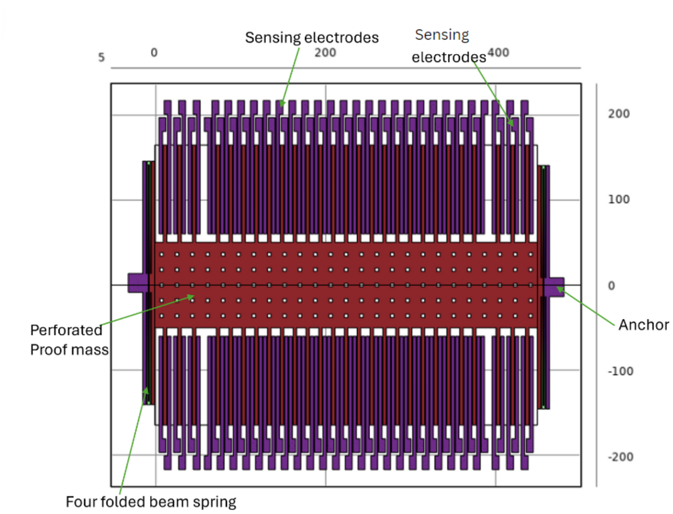

Fig. 3. The differential capacitance accelerometer original design from USN (All units are in micrometer)

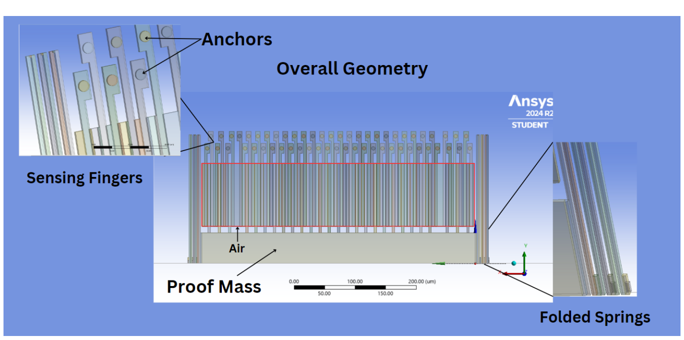

Fig. 4. The Accelerometer Sensor Geometry in ANSYS

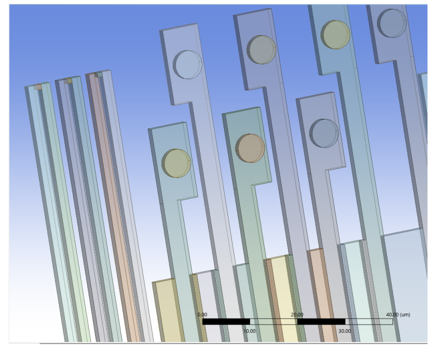

Fig. 6. Zoomed View of the Springs and Fingers Structure

TABLE II
DIMENSIONS OF THE SENSOR GEOMETRY

Parameters	Value
Proof of mass Length	448 µm
Proof of mass Width	100 µm
Thickness of Proof mass	2 µm
Air Strips Width	1 µm
Air Strips Length	104 µm
Thickness of Air Domain	2 µm
No of Sensing fingers	54
Length of fingers	114 µm
Width of fingers	4 µm
Capacitance gap	1 µm
Spring beam Length	210 µm
Spring beam Width	2 µm

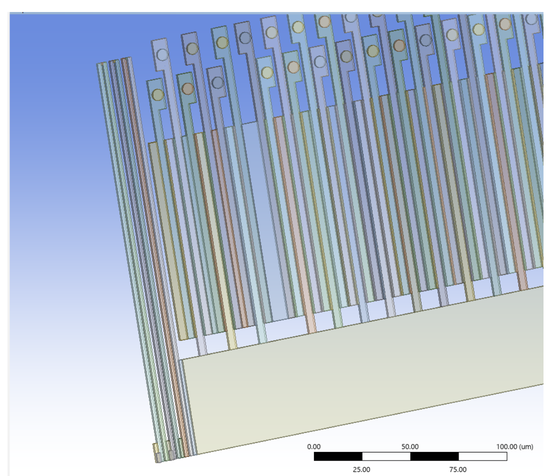

Fig. 5. Structure of the Springs and the Fingers attached to the Proof Mass

D. Meshing

The overall meshing gemetry is shown in Figure 7. The meshing details of the various components of the model are shown in Figure 8 and Figure 9. A conformal mesh was applied to the air domain, ensuring smooth transitions between regions. The proof mass was meshed using a tetrahedron mesh, while the springs were meshed using edge sizing and sweep methods. A multizone method was used for the fingers, with face and edge sizing. The anchors were meshed using a sweep method, combined with edge sizing.

Table III summarizes the global mesh settings and statistics, with the mesh resulting in a total of 974,119 nodes and 866,843 elements.

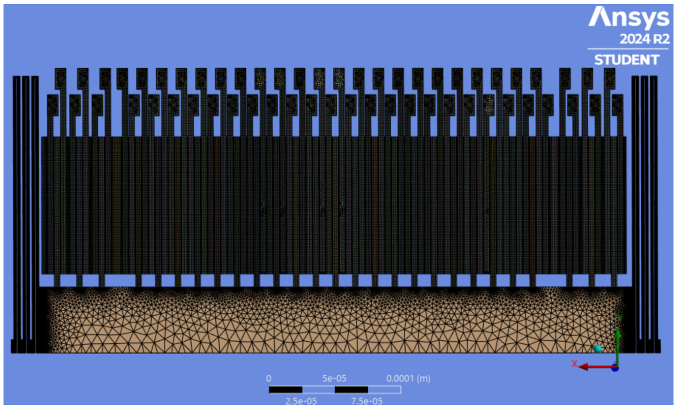

Fig. 7. Overview of the Mesh Geometry

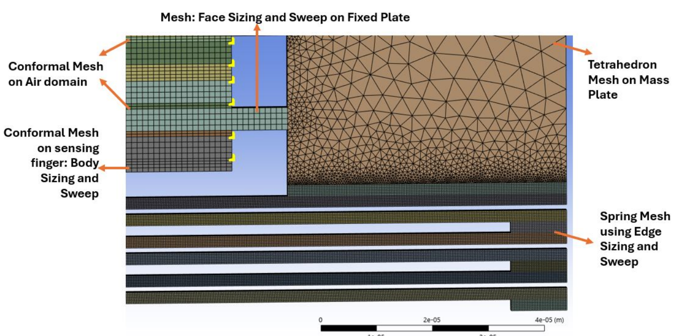

Fig. 8. Details of Mesh used on Proof mass, Air domain and Springs

TABLE III
GLOBAL MESH PARAMETERS & STATISTICS

Mesh Parameters	Values/Settings
Elements size	2.695e-05 m
Growth Rate	1.2
Physical Preference	Nonlinear Mechanical
Element Order	Linear
Nodes Number	974119
Element Number	866843

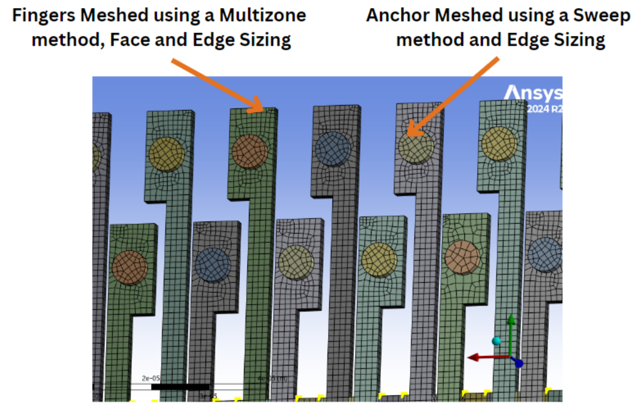

Fig. 9. Details of Mesh used on Fingers and Anchors

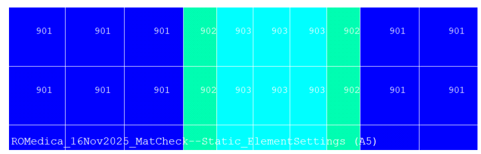

Fig. 10. Element Type Setting for Geometry

E. Test load simulation

The device functions as comb finger types that represent the parallel plate capacitor when the biased voltage is applied. The electric field is generated and energy is extracted to provide the change or differential capacitance between two electrodes. The realization is done through coupled simulation that introduces the effect of electrostatic based on biased voltage and applied load. The coupled field elements are selected through element type Solid226 with keyoption (1001) which represents the coupled electrostatic structural analysis. The biased voltage of (+1.5, -1.5) is applied on fixed electrodes and (0 V) on moveable electrodes. The air-solid boundary is defined through key option (4,1) and the remaining air domain is configured with no electrostatic forces. Thus, three element types are defined for solid, air, and air-solid boundary as shown in Figure 10.

Once the element settings are modified, the boundary conditions are defined as fixed electrodes and fixed faces for spring as also defined in Figure 3 so that structure is deflected only in X–axis direction.

The solver settings for static study are done using resulting output using electric field analysis by enabling large deflection using *nlgeom* and further controls are defined through the interactive solution settings sparse solver with multiple mesh layers configurations. The most important setting is done for capacitance calculations using *cmatrix* APDL command. The explanation by [4] is followed to use the three conductors. For our case, these conductors are conductor-1, and conductor-2 for small and large fixed fingers respectively and conductor-3 is grounded (moveable finger). The electrostatic domain is considered only for calculating the capacitance values, and solid elements were dropped.

The coupled simulation study investigation is done through two phases. In phase one, the effect of input bias voltage

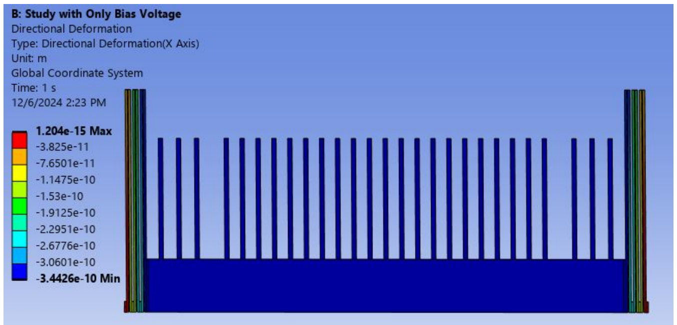

Fig. 11. Biased Voltage Result - Without Acceleration Load

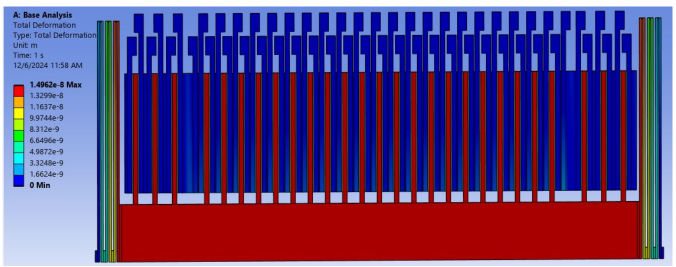

Fig. 12. Deflection of Device at 1g with Biased Voltage

is simulated without any additional load to detect the default deflection of the accelerometer. The result mentioned in image Figure 11 indicates that default deflection of the moveable structure is almost zero at biased voltage. Therefore, deflection contributions only come from acceleration load. Table IV presents the simulated vs analytical values. In the second phase, acceleration load with bias voltage enabled is simulated to see the deflection of the device. The overall device deflection is found to be 14.27 nm at 1g acceleration load which matches with the previous simulation study done in COMSOL multiphysics modeling. The result of this deflection study is shown in Figure 12. The intended acceleration range for this device is -4g to 4g and deflections are extracted for all ranges with step size of 0.5 (from 0.5 to 4g). The resulting analytical calculations and simulated values for deflection are presented in Table IV. The analytical displacement values are calculated using the equation 6 78.

TABLE IV
COMPARISON OF SIMULATION AND ANALYTICAL DISPLACEMENT

Acceleration (g)	Displacement_X Simulation (m)	Displacement_X Analytical (m)	Difference (%)
4.0	5.7433×10^{-8}	5.2891×10^{-8}	8%
3.5	5.0360×10^{-8}	4.6280×10^{-8}	8%
3.0	4.3230×10^{-8}	3.9669×10^{-8}	8%
2.5	3.6050×10^{-8}	3.3057×10^{-8}	8%
2.0	2.8821×10^{-8}	2.6446×10^{-8}	8%
1.5	2.1562×10^{-8}	1.9834×10^{-8}	8%
1.0	1.4275×10^{-8}	1.3223×10^{-8}	7%
0.5	6.9680×10^{-9}	6.6114×10^{-9}	5%

Under the bias voltage, the electrostatic force presents in the operation mode and the mechanical load is the acceleration. With only mechanical load the effective spring constant of the system is governed by the mechanical spring constant which can be calculated using Eq.10. The bias voltage induced electrostatic field generates a force which influences the net force acting on movable plate in a parallel plate capacitor as described in Eq.13. In our design the movable fingers attached to the proof mass will be under the influence of attraction force on either side of it and the resultant of it (Fe) contribute to the net force, this is described in Eq.14. The total force on body can be calculated using Eq.15 where F_{sum} is total force and F_{m}, F_{e} are mechanical and electrostatic forces respectively. The effective spring constant, based on spring softening effect in the presence of electrostatic force can be derived as in Eqn.16.

$$F_e = \frac{-U^2 \epsilon_0 \epsilon_r A}{2d^2} \tag{13}$$

$$F_{sum} = F_m + U^2 \epsilon_0 \epsilon_r A [\frac{1}{2(d-x)^2} - \frac{1}{2(d+x)^2}] \tag{14}$$

$$F_{sum} = F_m + F_e \tag{15}$$

$$\begin{aligned} K_{sum} &= \frac{dF_{sum}}{d_x} \\ &= K + \frac{dF_e}{dF_x} \end{aligned} \tag{16}$$

F. Output quantity dependence on deflection

The relation between output deflections achieved through parametric sweep of range (-4, 4 g). Unlike energy extraction method, the comb-driven capacitance is extracted through cmatrix [4] for each input load and ANSYS Workbench generated Cmatrix values are computed based on name selection. The Cmatrix is calculated based on two steps. First simulation is done coupled electrostatic simulation with acceleration load and given biased voltage on required conductors and it's without capacitance calculations. Then the deformed mesh of first step simulation is fed to second simulation and the calculation of the capacitance happens there. In a similar fashion, the resulted deflection from analytical and simulated studies was used to calculate and compute capacitance respectively as shown in Table V. The analytical capacitance calculations are done using equation 4.

The comparison of electrostatic forces and spring softening effect is presented in Figure 13 to showcase how the impact of spring softening can increase the deflection and it relates to bias voltages being applied to the device. For each case the resulting simulated values mentioned in Figure 14 15

The polynomial order is used to produce the equations. The Graph 14 and Graph 15 is obtained through calculated results presented in Table.V and Table.IV which shows the visual representation of the fitted curve with equations.

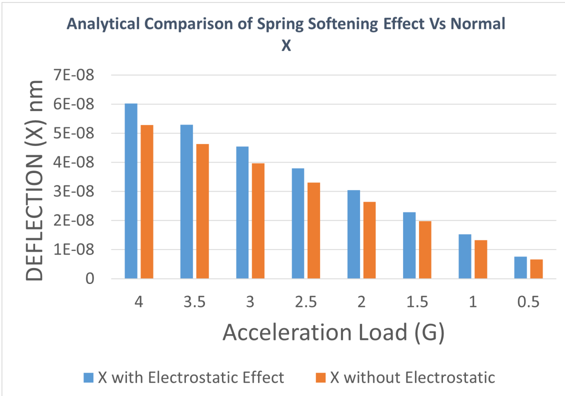

Fig. 13. Comparison of Deflection Values with Spring Softening Effect and without biased Voltages in Analytical Calculations

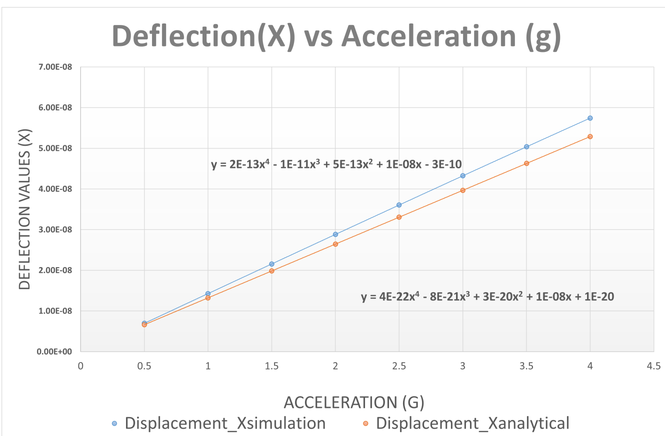

Fig. 14. Deflection Vs Acceleration Fitted Curve

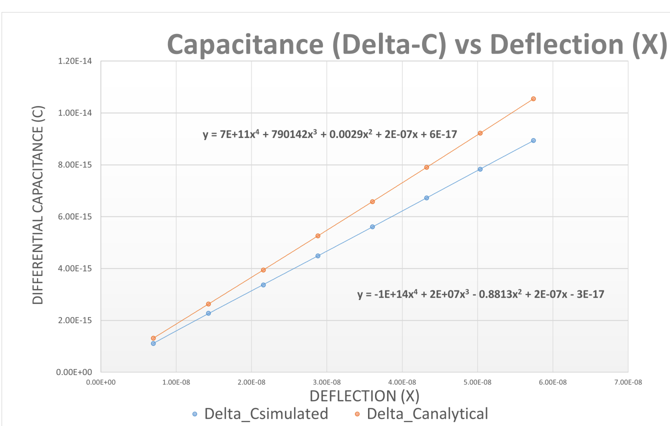

Fig. 15. Capacitance Vs Deflection Fitted Curve

TABLE V
COMPARISON OF SIMULATED AND ANALYTICAL CAPACITANCE CHANGES WITHOUT SPRING SOFTENING EFFECT

Acceleration (g)	**Delta_C Simulated (F)**	**Delta_C Analytical (F)**	**Difference (%)**
4.0	8.9360×10^{-15}	1.0547×10^{-14}	15%
3.5	7.8310×10^{-15}	9.2229×10^{-15}	15%
3.0	6.7240×10^{-15}	7.9008×10^{-15}	15%
2.5	5.6130×10^{-15}	6.5809×10^{-15}	15%
2.0	4.4900×10^{-15}	5.2626×10^{-15}	15%
1.5	3.3650×10^{-15}	3.9458×10^{-15}	15%
1.0	2.2750×10^{-15}	2.6299×10^{-15}	13%
0.5	1.1110×10^{-15}	1.3148×10^{-15}	15%

G. Sensitivity analysis

Sensitivity of the differential capacitance accelerometer can be measured using different metrics, with respect to change in displacement or capacitance or voltage against unit acceleration in terms of g.

In publications, the simulation-based claims presented the sensitivity in-terms of change in displacement/g and capacitance/g. TableVI list down the values.

TABLE VI
THE COMPARISON OF SENSITIVITY VALUES FROM DIFFERENTIAL CAPACITIVE ACCELEROMETERS PRESENTED IN LITERATURE

Author [ref] Year	Displ/g $\mathrm{nm\,g^{-1}}$	cap change/g $\mathrm{fF\,g^{-1}}$	volt change/g $\mathrm{mV\,g^{-1}}$
Chawla et al.[[5]] 2015	0.154	0.00785	-
Dwivedi et al.[[6]] 2020	-	5.91	-
Ghasemi et al.[[7]] 2024	-	-	48.51
Yuan et al.[[8]] 2015	-	1.44	-

The mechanical sensitivity in terms of displacement/g is calculated analytically using the calculated spring constant 10 and estimated proof mass. The calculated spring constant (K) without electrostatic softening effect is $0.2396\ \mathrm{N\,m^{-1}}$ and with electrostatic force is $0.2081\ \mathrm{N\,m^{-1}}$. Based on this the analytically calculated sensitivity is $14.2\ \mathrm{nm\,g^{-1}}$ which is better than the sensitivity achieved in literature listed in Table.VI. The differential capacitance calculated using *cmatrix* is listed in Table.V and it is in the order of fF across differnt load condition and it is comaparable with literature values of between $7\mathrm{aF\,g^{-1}}$ to $5.91\mathrm{fF\,g^{-1}}$. The sensitivity reached after amplification with our simulation is around $10\mathrm{mV\,g^{-1}}$ and it is in the same order of magnitude as recorded in the literature for similar application and die size [7].

H. Modal analysis

The modal analysis is performed with linear perturbation which is necessary for coupled field analysis. The numbers of maximum modes are found to be 10 as shown in Table VII, where because of low frequency requirements [9] one mode sufficed the requirements of our application. The modal analysis is done by applying the fixed boundary conditions from static analysis and suppressing the interior air degrees of freedom (DoFs). The modal frequencies are seen through Ansys mechanical but mode shapes are analyzed through APDL mechanical. This mode first natural frequency is well outside our required range of monitoring patients' activity during rehabilitation. The research suggests [1], [9] to measure the low-frequency components to track the progress effectively. The relevant mode shapes are shown in Figure 16.

TABLE VII
LIST OF MODAL FREQUENCIES OBTAINED UNDER LINEAR PERTURBATION MODAL ANALYSIS

Mode	**Frequency** (Hz)
1	4680.1545
2	7378.0032
3	33688.3689
4	34313.2208
5	34623.6411
6	34773.1385
7	38221.7930
8	38316.3496
9	38383.1563
10	38439.2363

I. ROM generation

The ROM generation is done using Ansys Workbench after selecting the number of desired nodes for deflection. The choice of node selection is based on movement of mass plate and fixed electrodes attached to it. The output node "m1" is shown in Figure 17 is at the center of a mass plate and "m2" is at the extreme right edge of the mass plate because the load accelerates the body in positive x-direction where "m2" displaces in that direction too. The node location is shown in Figure 18. Several node selection scheme is tested for example, the right, and the left edge of mass plate, face perpendicular to x-axis, and nodes on electrodes. All of them show the almost same displacement for the chosen model of accelerometer. The baseline study is generated at "1g" and sweeped from desired range of (-4,4 g).

J. ROM usage

The output displacements of the ROM nodes are mapped into voltage change (such as $\mathrm{fF\,g^{-1}}$ sensitivity to $\mathrm{\mu V\,g^{-1}}$) using curve fitting and analog circuitry respectively. The Figure 19 shows the response of the ROM as in setup 17 for the input load of 1 (1g), where applied load generated the output of around 15 nm in deflection which is equal to deflection obtained in ANSYS Workbench Figure 12. The frequency response also calculated by extracting the time duration required for 10 periods (between m2 and m3 markers) and it provides 4651 Hz, which is closer to the first modal frequency of 4680 Hz. It is observed that the setting parameters like transients simulation setting, and trapezoidal load input is critical to view the output data. For example, in our case, rise time is 5ms which represent the system reach to the 50% of the final value which is around 7.5nm.

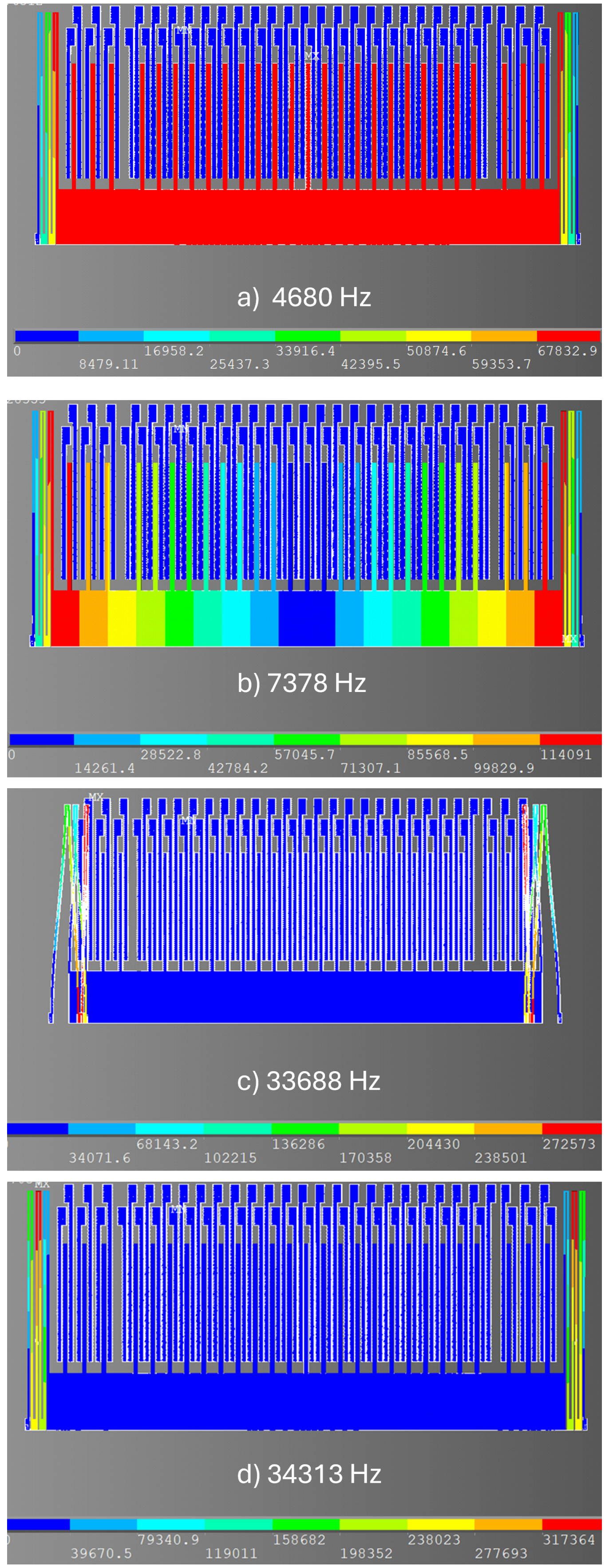

Fig. 16. (a-d) First, Second, Third & Fourth mode shapes respectively

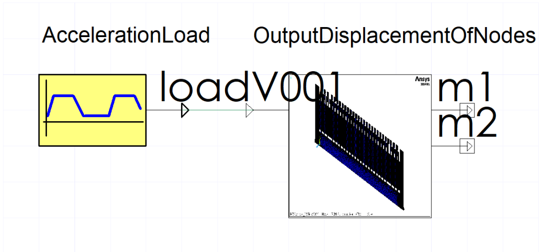

Fig. 17. ROM For Twin Builder - Accelerometer

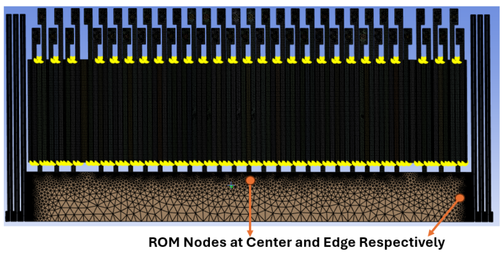

Fig. 18. Location of Chosen Nodes for ROM

The displacement received for each acceleration load with FEM simulation is compared with the displacement output of the ROM against respective acceleration load. The plotted results in 20 shows that the ROM outputs are aligned with the FEM simulated result.

K. Conclusion of the section

The accelerometer's reduced order modeling presented an optimized way to analyze the device behavior under subjected load conditions. The biased voltage plays a key role in extracting the required capacitances from electrodes. The division of geometry fixed and movable electrodes into three

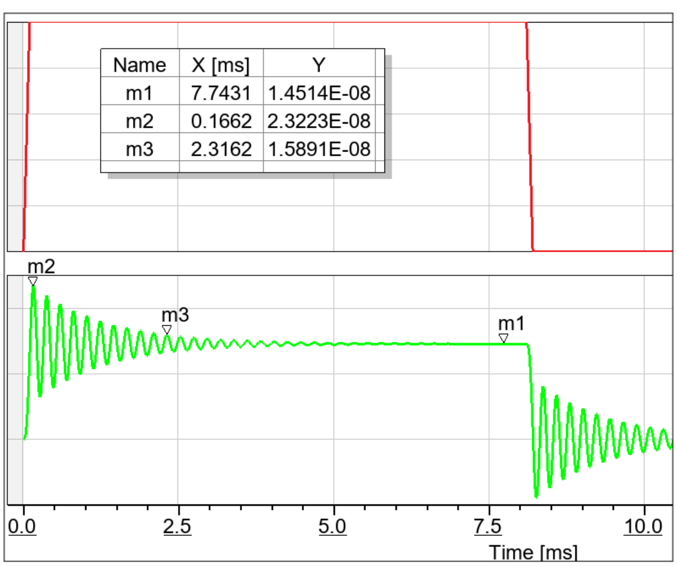

Fig. 19. Response of ROM for the acceleration load of 1g

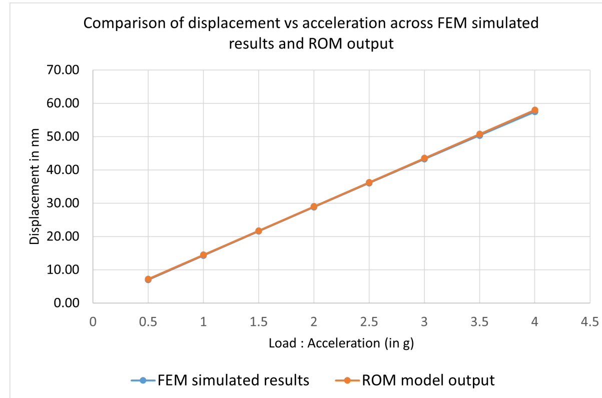

Fig. 20. Comparison of displacement obtained with FEM simulation and ROM model against different acceleration conditions

different conductors has facilitated to compute the differential capacitances in terms of matrices. The simulated values are relatively close to the calculated capacitances, as in this complex geometry factors like meshing and FEM approximation model leaves a room for marginal difference. The biasing condition for this device has not affected the resonant frequency of the device which is around 4kHz and has been tested with and without biased simulation schemes. For analysis purposes and reducing the complexity, the designed geometry is simulated with symmetry boundary conditions in x-axis direction and presents the same deflection as compared to previous study computed in COMSOL software. The relative margin of the device between analytical and simulated values for deflection is less than 10% even with approximations. This resulted in targeted capacitance change which matches well with the output sensitivity change of the device in $\mathrm{mV\,g^{-1}}$.

IV. Analog Circuit Design

An analog circuit is required to convert, read and co-relate the capacitance change occurring due to acceleration on the accelerometer system. The circuit can be broadly understood using the block diagram shown in Fig. 21. It consists of three major stages - capacitance to voltage conversion; amplification and peak detection. The circuit shown in Fig. 22 is designed for this. It consists of the changing capacitor C_s and C_r (the two capacitors in the differential capacitance). Two voltage sources with a phase difference of 90 degrees drive the two capacitors (i.e. electrodes). The C_p here stands for the capacitance due to parasitics and C_i is the input capacitance of the op-amp stage. Then this voltage is taken through C_s and C_r and brought to op-amp to be amplified. As the signal is amplified, it gives a similar output as the input but with an amplified amplitude. Since our driving was a sinusoidal input, it gave us a sinusoidal output. This is then given to a precision peak detector circuit, which feeds a DC voltage to a voltage-controlled oscillator to generate a frequency corresponding to the voltage input. This output frequency can be then measured by a digital circuit to display on a 7-segment display.

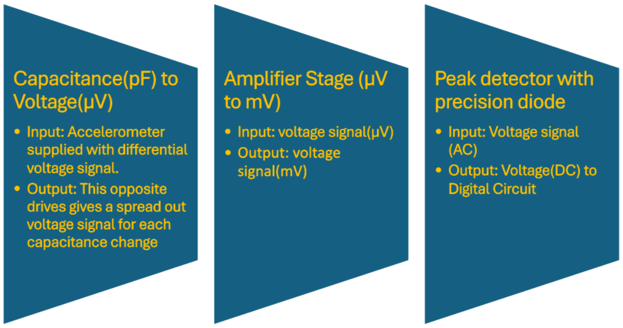

Fig. 21. Analog Circuit Block Diagram

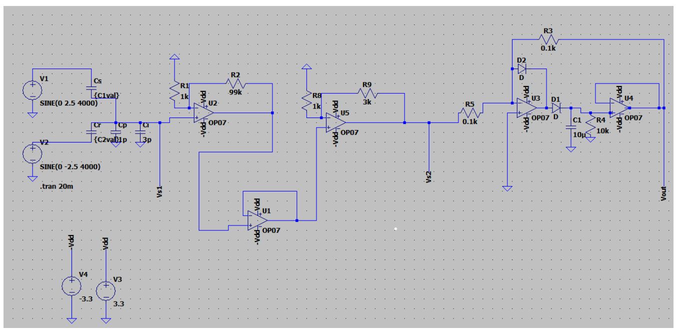

Fig. 22. Schematic for Capacitance to Voltage Conversion

The simulation is performed by changing the change in capacitance (C_r - C_s) for each value obtained from acceleration of 0.5g to 4g with a step of 0.5g. The input voltage is sinusoidal at ±2.5V with a frequency of 5KHz, which is kept to be greater than the first modal frequency of the accelerometer. The output of the first stage can be understood by the following equation.

$$V_{s1} = \frac{(C_s - C_r) \cdot V_r}{C_s + C_r + C_i + C_p} \tag{17}$$

where V_{s1} is the output of the first stage, as shown in Fig. 23. After this, this minute voltage V_{s1} is converted to an amplified

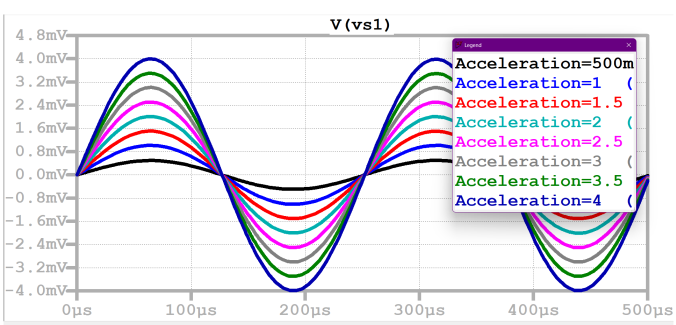

Fig. 23. Capacitance to voltage from the first stage.

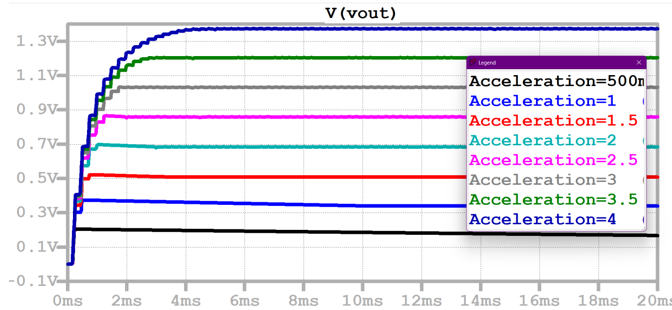

Fig. 24. Simulation Output of Voltage at the final output stage with different capacitance values.

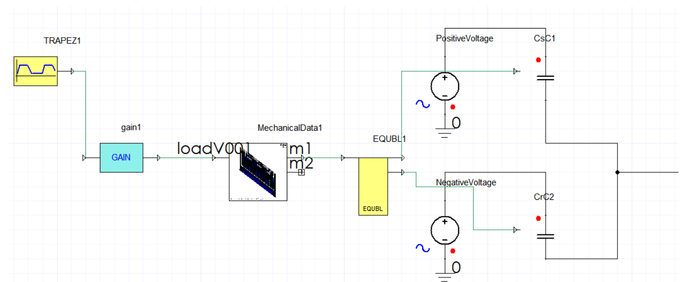

Fig. 25. System to develop capacitance from ROM with input as acceleration

value using an amplifier. We get different discernible voltages V_{s2} for each of the differences in capacitances. According to the output voltage requirements, the amplification factor (A) can be increased more as required by adding op-amp stages.

$$V_{s2} = A \cdot V_{s1} \tag{18}$$

It is then fed to a peak detector to convert the AC signal V_{s2} to a DC V_{s3} to be transmitted to a frequency measurement circuit.

$$V_{s3} = max[V_{s2}]) \tag{19}$$

The current output voltage varies in the order of 1 V for change in the capacitance of 4g and should be good to drive a frequency change. This voltage can be seen in Fig. 24 along with the changing acceleration values.

This circuit is used along with the ROM Model as shown in Fig. 25 where the input is acceleration and the output is the displacement of the mass and the fingers inside the sensor. This changes the capacitance. The change in capacitance with respect to the displacement is modelled in Twin Builder using an equation block as shown in Fig.25. The discussed analog circuit is used to decipher the capacitance change to voltage change which drives the voltage-controlled oscillator. This would give a required frequency in a desired range to be fed into the frequency measurement circuit. The simulations were run for different values of acceleration (see VIII) to find out the sensitivity which turned out to be 1mV/g which is then translated to 1V/g with an amplification factor of 1000.

TABLE VIII
VOLTAGE VALUES AT STAGE 1 OF THE CIRCUIT FOR DIFFERENT G VALUES

Acceleration (g)	Displacement(m)	Cl(F)	C2(F)	Vs1(V)
4	5.7433E-08	1.0418E-13	9.5244E-14	4.230000E-03
3	4.323E-08	1.0293E-13	9.6206E-14	3.220000E-03
2	2.8821E-08	1.017100E-13	9.722E-14	2.160000E-03
1	1.4275E-08	1.0054E-13	9.826500E-14	1.092E-03

As the overall circuit is designed, the individual component of the Operational Amplifier is very crucial and has to be designed keeping in mind the following specifications to amplify the 1mV/g signal.

TABLE IX
KEY SPECIFICATIONS OF THE OPERATIONAL AMPLIFIER

Parameter	**Value**	**Units**
Open-Loop Gain (AFB)	1000	-
Total-Loop Gain (0.1% error)	1000000	-
Gain-Bandwidth Product (GBW)	5	MHz
Maximum Operating Frequency	5	KHz
Voltage Swing (Vpp)	3.3	V
Output Offset Voltage (Voffset_out)	0.1	V
Phase Margin	30	degrees
Load Capacitance (CL)	1 pF	F

A. *Comparison of topologies*

Different amplifier configurations offer specifications which is suitable for a specific application but not applicable for another, thus it is necessary to prioritize one feature over other out of gain, linearity, supply voltage, voltage swings, speed, input and output impedance, power dissipation and noise. The operational amplifier is realized off adhering to the ideal characteristics of infinite voltage gain, infinite input impedance, zero output impedance and infinite output current as closely as possible. The common topologies available are single-stage topology, two-stage topology, telescopic cascode topology, folded cascode topology and regulated cascode topology.

1) Single Stage Topology: The single-stage amplification can be divided into three based on which terminal is common to both input and output Common Source, Common Drain and Common Gate. Each of these configurations has unique characteristics and is thus suitable for applications which require such features. Common Source offers high gain and is used in general-purpose applications, while Common Drain offers high input impedance and low output impedance with unit gain and thus acts as a voltage follower. Common Gate configuration provides low input impedance and high output impedance and is widely used in high-frequency applications.

2) Two Stage Topology: The two-stage amplifier is applicable when significant gain is required, where two amplification stages are connected in series to achieve high gain and high voltage swing. The first stage is a differential pair, where the load is usually active, and the second stage is a common source amplifier. The two-stage topology provides high gain and better control over the input and output impedance matching and bandwidth [10]. Furthermore, the distribution of gain across two stages effectively lowers the noise level and provides better design flexibility.

3) Telescopic Cascode Topology: The telescopic cascode provides high voltage gain and improves the bandwidth by reducing the Miller effect. The differential pair amplifies the input signal, and the drain terminal of each transistor is connected to a cascode transistor and then connected to the power supply [11]. The steps of transistors create a compound lens structure visually and thus a “telescopic” structure. This topology is suitable for low-noise applications and the complex biasing, limited output swing is the drawback.

4) Folded Cascode Topology: The limited output voltage swing in the telescopic cascode is rectified by folding the current path. It is suitable for low voltage operations as it does not stack the multiple transistors across the supply terminals and on the other hand it consumes increased power due to the additional circuitry. It has good frequency response and better linearity and is thus suitable for high-precision analog circuits [12].

5) Regulated Cascode Topology: This is the advanced version of cascode topologies, where an additional amplifier is included to regulate the voltage of the cascode transistor's gate. The performance is improved in terms of gain and linearity at the expense of increased complexity and high-power consumption.

The comparison of different topologies is presented in Table X.

TABLE X
COMPARISON OF DIFFERENT AMPLIFIER TOPOLOGIES

Topology	**Gain**	**Bandwidth**	**Speed**	**Output Swing**
Two Stage	High	High	Low	Highest
Telescopic cascode	Medium	Medium	Highest	Medium
Folded cascode	Medium	Medium	High	Medium
Regulated cascode	High	High	Medium	Medium

B. *Description of Selected Topology*

To achieve a closed-loop gain of 1000 (60 dB), the op-amp must have a sufficiently high open-loop gain to minimize gain error and maintain stability. For a gain error of 0.1%, the open-loop gain needs to be 1,000,000 (120 dB). This is here realized using a two-stage op-amp: a high-gain first stage, followed by a second stage for further amplification and driving the load and a source-degenerated pre-amplifier stage providing an initial gain of 20 dB. This will improve linearity and input matching with a high-gain main amplifier stage to achieve the required total open-loop gain of 120 dB.

Firstly to achieve a gain of 20dB, we use a source degeneration amplifier as shown in Fig. 26

This is an amplifier where a resistor is placed in series with the source connecting two MOSFETs. This resistor reduces gain slightly but improves linearity, input impedance, and stability. It is used before a two-stage op-amp to provide an initial gain of 20 dB because it amplifies the input signal modestly with good linearity. It also improves input impedance

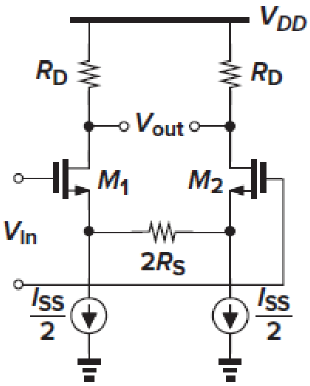

Fig. 26. Source Degeneration applied to the differential pair.

for better signal matching, and reduces distortion for the main op-amp stages. Its gain is given by the formula

$$A_v = -G_m R_D = \frac{-g_m R_D}{1 + g_m R_S} \tag{20}$$

As evident the values of R_D and R_S are set to obtain a gain of 20dB. After this, the selected two-stage op-amp stage is used to obtain the required further gain of 100dB. It consists of a differential amplifier a gain stage followed by a voltage-controlled voltage source (VCVS) as the output stage. The differential stage amplifies the difference between two input signals. It also gives a high common-mode rejection ratio. The gain of this two-stage amplifier is expressed as:

$$A_v = A_1 . A_2 \tag{21}$$

where the A_1 and A_2 are the gain of the individual stages of the two-stage op-amp. The output from this second stage is fed into a VCVS allowing for handling the load stage.

C. Hand Calculations

This section deals with the analytical calculations of two-stage amplifier and source degeneration amplifier. Starting with the compensation capacitance (C_c), it balances stability and bandwidth by addressing pole-splitting in the amplifier. The slew rate (SR) and tail current (I_{tail}) govern the amplifier's ability to respond to rapid input signal changes, ensuring fast and accurate output. Transconductance (g_m) and aspect ratios (W/L) define the sizing of transistors to achieve the desired gain, while gain-bandwidth product (GBW) directly impacts the amplifier's frequency response.

Stability metrics, including the dominant pole (f_1), non-dominant pole (f_2), unity-gain bandwidth (UGB), and phase margin (ϕ_m), are crucial for preventing oscillations and achieving a robust system with proper phase response. Together, these equations show the trade-offs in speed, gain, power consumption, and stability critical for high-performance analog circuit design. The input parameters for these calculations are summarized in Table XI.

The analytical formulas used for defining the parameters are summarized in Table XII. It is important to note that to match the tail current that for (M_8) transistor, the scaling factor of (3.5) is used.

TABLE XI
INPUT PARAMETERS FOR THE DESIGN

Parameter	**Value**	**Description**
C_L	10 pF	Load capacitance
f_{max}	5 MHz	Maximum operating frequency
V_{pp}	3.3 V	Peak-to-peak voltage
V_{on}	50 mV	Overdrive voltage
V_{thn}	0.50 V	NMOS threshold voltage
V_{thp}	0.65 V	PMOS threshold voltage
K_n	170 μA/V^2	NMOS process constant
K_p	60 μA/V^2	PMOS process constant
λ	0.05	Channel length modulation parameter
A_{FB}	10^6	Feedback gain (120 dB)
$V_{\text{offset, out}}$	0.1 V	Output offset voltage
A_{Vth}	9.5 mV $\cdot$ μm	Matching parameter

TABLE XII
KEY EQUATIONS FOR TWO-STAGE OP-AMP DESIGN

Equation	**Unit/Value**
$C_c = 0.2 \cdot C_L$	2.0 pF
$SR = 2\pi f_{\text{max}} \cdot V_{\text{pp}}$	103.67 V/μs
$\text{V}_{\text{offset\_in}} = \frac{V_{\text{offset\_out}}}{A_{\text{FB}}}$	100 μV
$I_{\text{tail}} = 1.1 \cdot C_c \cdot SR$	207.34 μA
$g_m = \frac{2 \cdot (I_{\text{tail}}/2)}{V_{\text{on}}}$	4.146 mA/V
$\frac{W}{L}_{\text{diff}} = \frac{2 \cdot (I_{\text{tail}}/2)}{K_n \cdot V_{\text{on}}^2}$	488.0
$GBW = \frac{g_m}{C_c}$	2073.0 MHz
$W \cdot L = \left(\frac{A_{Vth}}{V_{\text{offset\_out}}/AFB \cdot \sqrt{2}} \right)^2$	4689.6
$\frac{W}{L}_{M_3} = \frac{2 \cdot (I_{\text{tail}}/2)}{K_p \cdot V_{\text{on3}}^2}$	154.0
$\frac{W}{L}_{M_7} = \frac{2 \cdot I_{\text{tail}}}{K_n \cdot V_{\text{on}}^2}$	488.
$\frac{W}{L}_{M_8} = 3.5 \cdot \frac{W}{L}_{M_7}$	1708.0
$r_{ds} = \frac{1}{\lambda \cdot I_{\text{tail}}}$	96.45 KΩ
$f_1 = \frac{1}{2\pi r_{ds} C_c}$	6.47 KHz
$f_2 = \frac{g_m}{2\pi C_c}$	518.36 MHz
$\text{UGB} = \frac{g_m}{2\pi C_c}$	330 MHz
$\phi_m = 90^\circ - \tan^{-1}\left(\frac{\text{UGB}}{f_1}\right) \times \frac{180}{\pi} - \tan^{-1}\left(\frac{f_2}{\text{UGB}}\right) \times \frac{180}{\pi}$	57.52 deg

In order to set the gain of the source degenerate amplifier, the values of R_D and R_S are chosen so as to produce a gain of 20dB. This is achieved by setting R_D as 8kΩ and $2R_S$ as 0.4kΩ. This should ideally give a gain of 20dB by using the formula in equation 20. The value of R_D is optimized to 9.28kΩ by performing AC simulation to find maximum gain at various values of resistances. The W/L ratio of the MOSFETS as set so as to flow appropriate current in accordance with the two stage circuit.

D. Cadence Schematics

According to the hand calculations, the schematics are designed using cadence. They use the AMS 0.65 µm and analoglib library to build the schematics. Further various

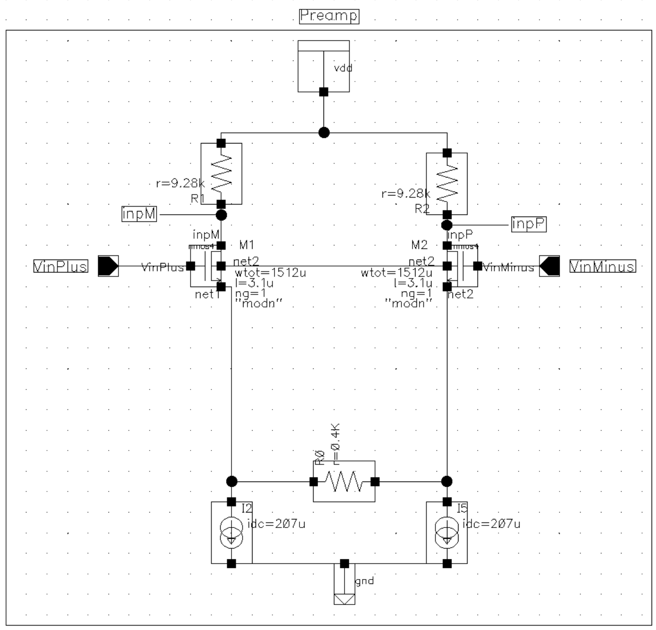

Fig. 27. The pre-amplifier stage

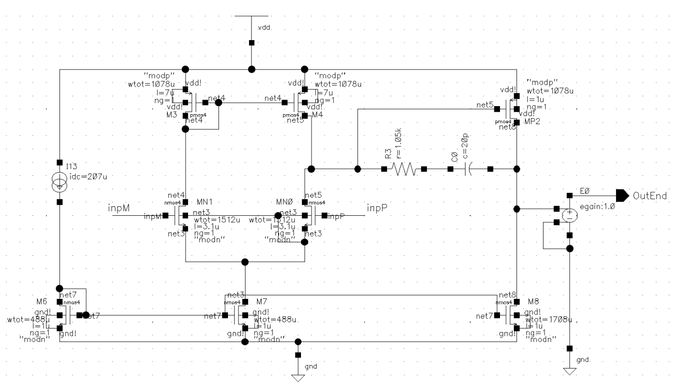

Fig. 28. The second stage - 2 Stage Differential Amplifier

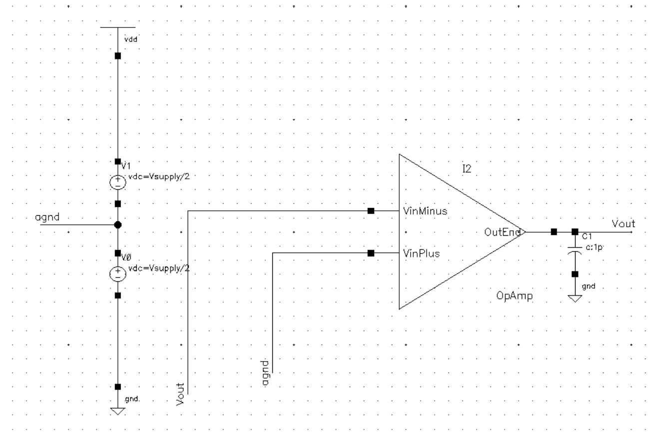

Fig. 29. Complete Op-Amp with a testbench

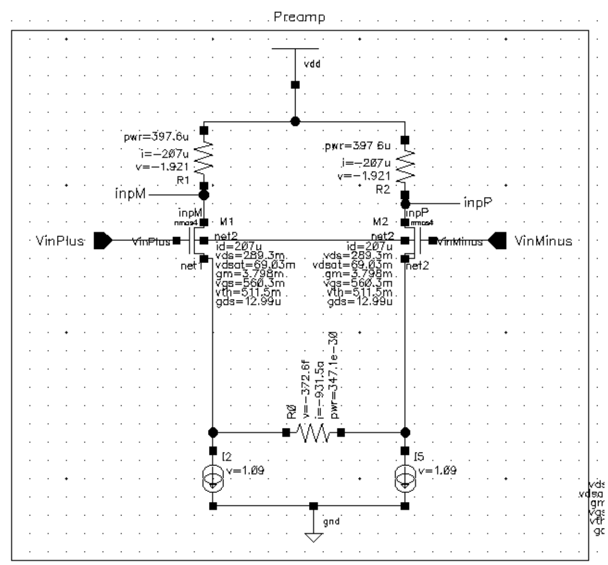

Fig. 30. DC Operating points for the MOSFETs

simulations such as DC, AC, Stability, Transient, Monte Carlo, and PVT analysis are performed to verify the hand calculations and the design. The Vdd is 3.3V and the ground is analog ground at 1.65V everywhere, except for the supply voltage. As shown in Fig. 27 is the pre-amp stage to enhance the input signal. Fig. 28 shows the two-stage amplifier acting as the main gain amplifier. The circuit in Fig. 29 depicts the complete op-amp as a symbol and the external testing circuit attached to test its gain.

E. DC Analysis

DC analysis is performed to find and set the operating points of all the MOSFETs so that the amplification can take place correctly in the saturation region. To perform this, DC voltage is applied at the gate terminals of the MOSFETs which must be greater than the threshold voltage. The drain to source voltage (V_{DS} must also be greater than the difference of gate to source and threshold voltage ($V_{DS} > V_{GS} - V_T$). At these conditions for the MOSFETs, the DC simulation is done to find the correct operating point to receive a Vdd/2 as the voltage output after the DC simulation. As shown in Fig. 30 and Fig. 31 are the DC operating points at which all the MOSFETs are working in order to give the desired amplification and stability during operation. The common mode voltage is given as 1.65V, and the differential as 0V.

F. AC Analysis

By performing AC analysis we can find the open loop gain of the amplifier. It is done by varying the input AC signal over a desired range of frequencies. Here we have simulated our design from 10Hz to 100MHz for the AC signal amplitude given at the non-inverting terminal to obtain the amplification

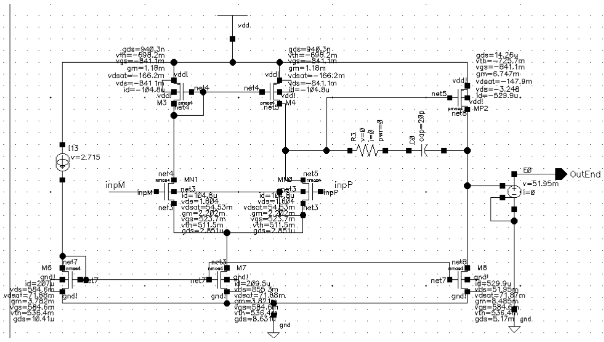

Fig. 31. DC Operating points for the MOSFETs

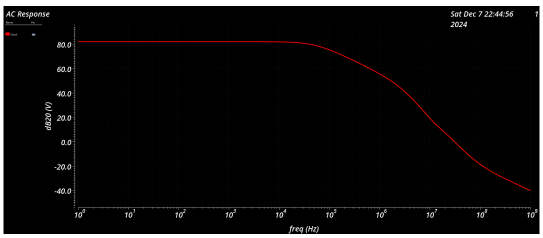

Fig. 32. AC response over the range of frequencies.

factor at all the frequencies. It gives a gain of around 80dB (see Fig. 32)for lower frequency but it drops with the increase in frequency. During this analysis, the DC operating points are kept the same to run the analysis, only the AC signal frequency is varied over the range. It gives us an insight into how the circuit behaves for the range of frequencies. We also know that the amplifier is unstable as it reaches -180 degrees of phase and there is no phase margin. This means it must be compensated for the phase margin to be stable.

G. Stability Analysis

After performing the AC analysis, we know that the amplifier is not stable as it reaches -180° of phase and there is no phase margin. In order to make the system stable we will need to take the first pole near to the lower frequencies and the second pole farther in the higher frequency band so that the system becomes stable as the phase margin increases. Our target for the phase margin is in the range greater than 30 degrees minimum but, we strive for a phase margin in the range of 45 degrees to make it better. In order to do this two elements, R_f and C_c, miller capacitance and resistance, are applied between the output of the first stage of the Two-stage amplifier to the output of the second stage of the Two-stage amplifier. As we increase the C_c value the gain at our required frequency of around 4KHz reduces hence we will have to optimize the gain and the phase margins. From the simulation, it is noted that at a C_c of 40pF, we obtain a phase margin of around 55 degrees (see Fig. 34) but the gain at 4KHz reduces significantly when compared to the C_c of 20pF 33), which still gives a good gain of around 45 degrees.

By performing the stability analysis, we find the loop gain with the feedback circuit. This is done by applying a zero

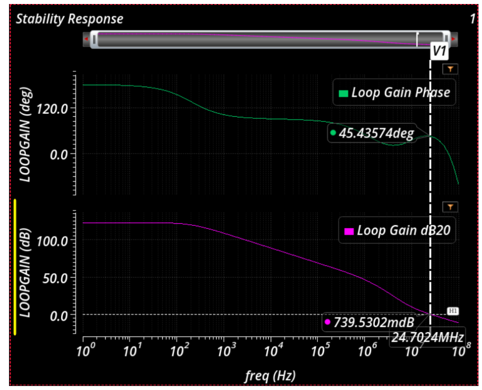

Fig. 33. Gain and Phase with C_c = 20pF

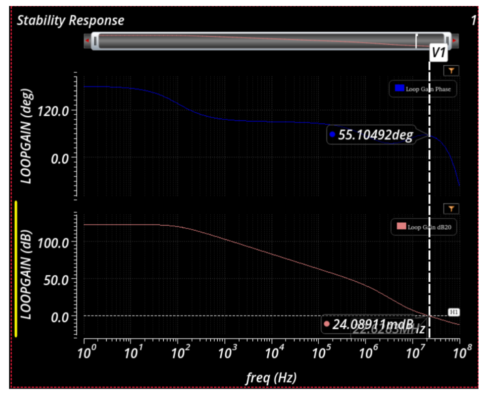

Fig. 34. Gain and Phase with C_c = 40pF

voltage source between the output and the inverting terminal, completing the feedback loop. The non-inverting terminal is applied with the DC and the AC signal voltage, which is varied over a desired range of frequencies. Here, we have simulated our design from 10Hz to 100MHz for the AC signal.

In Fig. 35 we can observe the gain of the amplifier circuit at 5KHz frequency. It is around 95 dB, and has not reduced as much as compared to when 40pF is used.

Another interesting sets of simulations were done to find

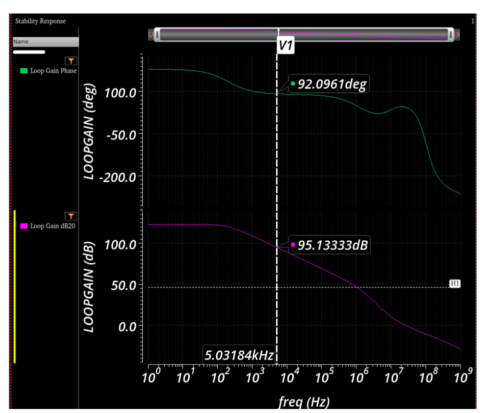

Fig. 35. Gain and Phase at 5kHz

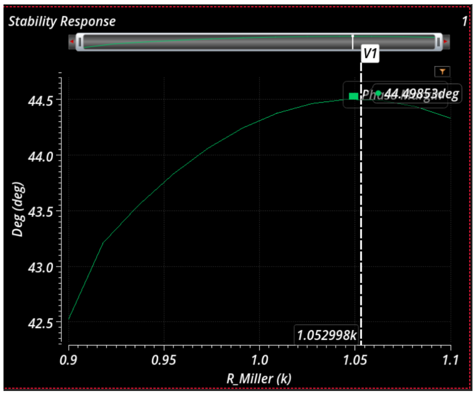

Fig. 36. Phase Margin vs R_f with C_c = 20pF

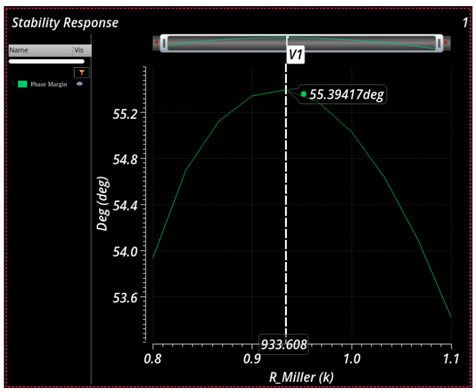

Fig. 37. Phase Margin vs R_f with C_c = 40pF

the optimum value of R_f the resistance used with the miller capacitances for both the values of 20pF and 40pF as shown in Fig. 36 and Fig. 37 respectively. It was done by performing a parametric sweep for the R_f and noting the phase margin for the circuit each time with the stability analysis configuration.

H. PVT Analysis

Choosing the right transistors or electronic components is very important because different manufacturers use various technologies. These differences can include things like the threshold voltage, channel length, oxide thickness, and materials used. The performance of an amplifier also changes with variations in supply voltage, which affects how much current flows and how much power is used. Temperature is another key factor; it determines how well the operational amplifier can work without issues.

To ensure that the amplifier operates reliably, these three factors must be tested for stability. This can be done using PVT (Process, Voltage, Temperature) analysis in Cadence. First, a stability analysis is conducted and the PVT corners are created in a.dcf file format. The temperatures selected for the analysis were -40°C, 27°C, and 85°C. The voltage varies at 3.1V, 3.3V and 3.5V for worst speed and worst power in cmos. In the ADE XL window, the stability analysis is set by adding the created corners and two test cases for C_c of 20pF and 40pF.

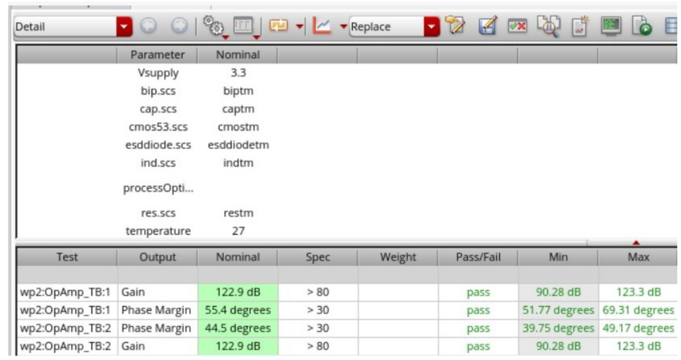

Fig. 38. PVT Test Validation in Cadence

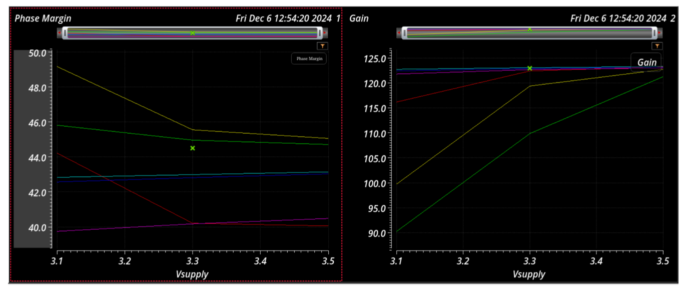

Fig. 39. Phase Margin and Gain variation with the changing supply and process parameters.

The specifications for phase margin and gain are set to a phase margin greater than 30 degrees and a gain greater than 80 dB.

After everything was set up, we ran the PVT analysis and received positive stability results. The results as shown in Table XIII demonstrate that all tests finished successfully, it can further be validated through Figure 38. Both TB:1 Gain and TB:2 Gain exceed the required threshold of 80 dB, with nominal values of 122.9 dB and a phase margin of 55.4 degrees and 44.5 respectively for general test corner at 27C. The TB:1 and TB:2 tests also surpassed the 30-degree specification for all the temperature, process and voltage variations.

TABLE XIII
PVT RESULTS

Test	Nominal	Spec	Pass/Fail	Min	Max	R_f	C_c
TB:1 Gain	122.9 dB	> 80	pass	90.28 dB	123.3 dB	933.33 Ω	40 pF
TB:1 Phase Margin	55.4 °	> 30	pass	51.77 °	69.31 °	933.33 Ω	40 pF
TB:2 Phase Margin	44.5 °	> 30	pass	39.75 °	49.17 °	1.05 kΩ	20 pF
TB:2 Gain	122.9 dB	> 80	pass	90.28 dB	123.3 dB	1.05 kΩ	20 pF

From the Fig. 39, it is evident that as the supply voltage increases the gain gets enhances but the phase reduces as compared to lower supply voltages.

From the Fig. 40, we note that as the temperature increases the gain of the op-amp reduces as it leaves the optimum operating point. The stability reduces as the temperature deviates from the room temperature.

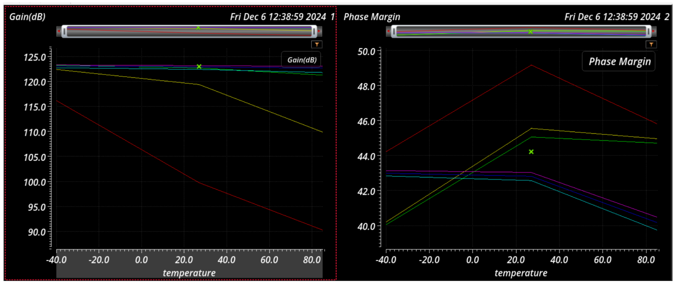

Fig. 40. Gain and Phase Margin variation with the changing temperature and process parameters.

I. Monte Carlo Simulation

Monte Carlo analysis is a statistical simulation technique employed to model variations in processes that influence the overall performance of VLSI analog designs. This method utilizes a series of random numbers to assess device mismatches and process variations. Multiple random samples are generated for the parameter vectors, which are then analyzed to evaluate the desired circuit's performance. The distribution of performance variables is typically represented using graphical statistical methods with histograms and scatter plots. Key statistical indicators derived from this analysis are the mean and standard deviation.

In our Monte Carlo analysis, we performed 100 iterations to simulate the complete circuit, focusing on both statistical variations and process mismatches related to the input differential pair and offset voltage. It is done by connecting the feedback loop. The inverting terminal is connected to the output of the op-amp and the input is given at the non-inverting terminal, making a voltage follower. For this the input is just the DC signal and the output is observed for varying parameter vectors to form a histogram. The histogram displaying the results from the Cadence simulation is illustrated in Fig 41, showing device performance and variation on the output offset voltage. The statistical outcomes for the offset voltage performance indicate a value with a mean of 1.65V and a skewness of 71.2381mV. This histogram effectively illustrates how circuit performance varies due to mismatches of the pre-amp and the differential input stage.

J. Transient Analysis

Transient analysis is done in order to find the variation in the output signal when the signal is given abruptly from the input terminal. This analysis enables us to measure the speed of the circuit as it can analyze the state of the signals before steady state reaches during the transient. For our design, we gave a input of stream of pulse of 15mV on voltage with a pulse width of 4ms at the non-inverting end of the amplifier which amplifies the signal two times. The rise and fall time of the input were kept as 10µs. The circuit's amplification factor is kept as 2 by keeping the ratio of the two resistors in non-inverting op-amp condition as unity. The output is clean without any ripples at this frequency of operation as seen in

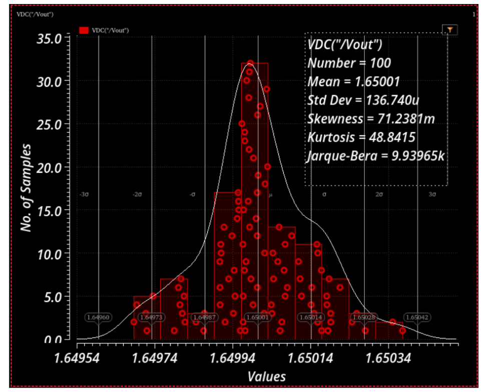

Fig. 41. Monte Carlo Analysis

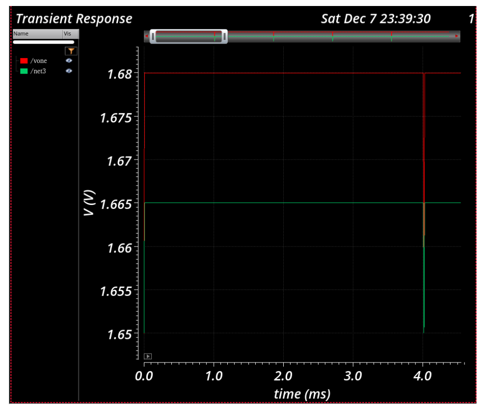

Fig. 42. Transient Simulation with pulse voltages

Fig. 42. The input is seen as 1.65+0.015V which increase to 1.65+0.03V due to amplifier's gain of two.

K. Amplification Test Circuit Simulation

DC simulation is done to simulate the amplification of the op-amp by varying the input voltage from zero to 15mV. To perform this, DC voltage is applied at the non-inverting terminal of the op-amp. As shown in Fig. 43 are the response of the amplifier. The amplifier is configured in non-inverting amplifier configuration. The resistor values are in the ratio of 99k/1k which gives a net amplification of 100, as visible in the figure. The amplification is given as:

$$\frac{V_o}{V_i} = 1 + \frac{R_2}{R_1} \tag{22}$$

Here the input of 15mV also becomes 30mV accordingly, verifying the correct operation of the designed op-amp.

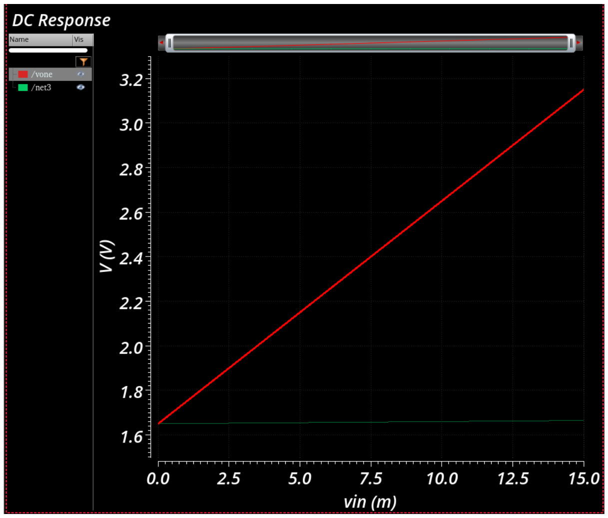

Fig. 43. Amplification using non-inverting op-amp configuration

Therefore, following rigorous hand calculations, schematic design and various simulations, the design is ready with the following set of specifications as shown in Table XIV

TABLE XIV
ACHIEVED SPECIFICATIONS OF THE OPERATIONAL AMPLIFIER

Parameter	Value
Gain Factor	122.9364 dB
Gain at 5 kHz	95 dB
Gain Bandwidth Product (GBW)	5000 kHz
Cutoff Frequency	207 Hz
Phase Margin	45°
Percentage Error (for 1mV signal)	0.1%
Input Offset Voltage	0.165m V
Output Referred Offset Voltage	1.65 V

V. DIGITAL DESIGN

A. *Digital Design Description*

The primary role of the digital design is to map the output voltage from the analog circuit into an acceleration value. It integrates a Voltage-Controlled Oscillator (VCO) as part of its overall architecture, along with a Frequency Measurement Circuit and an Acceleration Calculator. The voltage is passed into the VCO, which converts this analog voltage signal into a square wave signal whose frequency is proportional to the input voltage. The digital system then measures the output frequency over a defined time window, and feeds this measured frequency into the calculation circuit to compute the corresponding acceleration value. This overall concept design is illustrated in Figure 44.

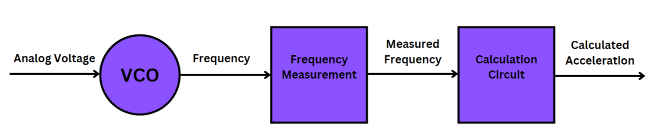

Fig. 44. Block Diagram of the overall Digital System

The digital system was designed using a hierarchical approach. The top-level module of the digital system, as illustrated in Figure 45, integrates two main sub-circuits: the Frequency Measurement Circuit and the Acceleration Calculator (Decision Circuit). This hierarchical approach was chosen to enhance design clarity, simplify debugging, and enable modularity while ensuring a clear separation of functionalities, allowing for easier future modifications.

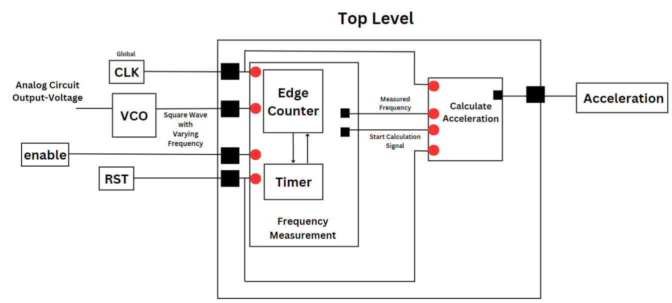

Fig. 45. Overall Digital logic indicating all subcircuits, Inputs and Outputs

The first component, Frequency Measurement Circuit, measures the frequency of the received incoming signal, it consists of two sub-modules: the Edge Counter and the Timer. The Edge Counter is responsible for counting the rising edges of the square wave signal generated by the VCO, while the Timer ensures that these edges are counted only within the specified time window, setting a time limit for measurement. The time window counter depends on the main clock frequency of the field-programmable gate array (FPGA), while the edge counter depends on the incoming signal frequency.

The frequency of the incoming signal is calculated based on the number of cycles (edges) counted within the specified time window, using the following equation 23:

$$\text{Frequency} = \frac{\text{Number of Edges}}{\text{Time\_Window}} \tag{23}$$

The second component, the Acceleration Calculator, processes the measured frequency from the Frequency Measurement Circuit. It computes the corresponding acceleration value using a the mathematical relationship in 24. This component functions only when it receives a signal (`start_calculation`) from the Frequency Measurement Circuit to start processing the incoming value, seamless operation between the two components.

$$\text{Acceleration} = \frac{F - F_0}{\text{sensitivity}} \tag{24}$$

where:

- F: Measured frequency.
- F_0: Nominal frequency.
- sensitivity: expressed in Hz/g. A product of K_{VCO} and the sensor sensitivity (V/g).

Furthermore, the top-level module introduces the main ports of the system: **CLK**, **RST**, **enable**, **freq_in**, and **acceleration**. The **CLK** signal provides the global clock to synchronize all components. The **RST** signal initializes the system, resetting all modules to their default states. The **enable** signal controls when the system begins functioning, serving as a simple activation mechanism. The **freq_in** signal serves as the input frequency from the VCO, while the **acceleration** output provides the computed acceleration value, based on the measured frequency.

The digital design was developed and verified using a comprehensive set of tools to ensure functionality, efficiency, and reliability throughout the design process. The system was modeled and implemented using VHSIC Hardware Description Language (VHDL), enabling the creation of a hierarchical digital structure. Functional simulations were conducted using ModelSim to ensure correct behavior of the design. Quartus Prime Software was employed for synthesizing the design, performing resource analysis, and timing analysis. Additionally, Twin Builder was used for system-level integration and simulation, ensuring seamless interaction between the analog and digital components.

Different reliability concerns are taken care of during the digital design process, starting from the synchronization between the Timer and Edge Counter state machines, which is ensured through the implementation of handshake protocols and acknowledgment signals. The Timer state machine uses the `counter_edge_ack` signal to indicate when the Edge Counter can begin counting, while the Edge Counter responds with the `start_counter_ack` signal to confirm readiness. This ensures that the counter will only start counting when the timer is active, and will stop when the timer stops. To handle clock domain crossings between the global clock (`clk`) and the input frequency (`freq_in`), double flip-flop synchronization is employed. This method passes the signals through two flip-flops, allowing them to stabilize before being read, thereby preventing metastability and asynchronous behavior. The process generates stable versions of the signals, `counter_edge_ack_stable` and `start_counter_ack_stable`, as illustrated in Figure 46.

Timing constraints, such as ensuring that the time_window can capture even the minimum (slowest) input frequency and that the system clock frequency is more than double the maximum input frequency, as required by the Nyquist-Shannon sampling theorem, are critical for achieving accurate frequency measurements.

Additionally, robust reset behavior is implemented through a `reset` port, ensuring that all signals and states are initialized predictably. These considerations collectively ensure that the

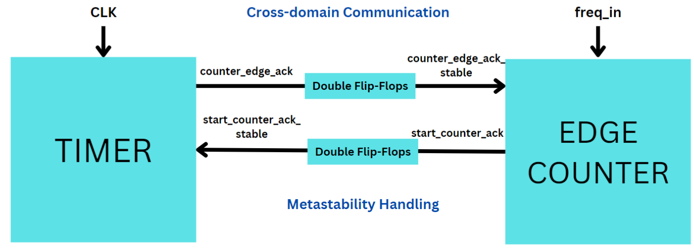

Fig. 46. Cross-Domain Communication with Metastability Handling for the Two Control Signals of the Frequency Measurement Circuit

design is reliable and capable of accurate performance under varying input conditions.

B. *Digital Design Implementation*

1) Design Parameters and VCO Configuration: The measurement time window was chosen to be 10 μs, as it meets the real-time measurement requirement of the application. The system operates with a global clock frequency of 50 MHz, which can be easily adjusted in the top-level design to accommodate different application requirements. The VCO element was implemented using VHDL-AMS (VHSIC Hardware Description Language - Analog and Mixed-Signal), with its output configured as a logical representation of '0' and '1'. The VCO parameters were set as outlined in Table XV.

The characteristic equation of the VCO is:

$$f_{out} = f_{nominal} + K_{VCO} \cdot V_{in}, \tag{25}$$

where f_{out} is the output frequency, $f_{nominal}$ is the nominal frequency (frequency at $V_{in} = 0$), K_{VCO} is the gain of the VCO, and V_{in} is the input voltage. The minimum frequency output from the VCO was adjusted to 10 MHz. This frequency lies above the minimum threshold that the selected time window can accurately accommodate, ensuring reliable operation. The nominal frequency, which is the frequency at $V_{in} = 0$, was set to 10 MHz. The maximum frequency of the VCO was set to 18 MHz, based on the expected range of the input signal, insuring that it is still less that double the clock, for accurate sampling. The value of K_{VCO} (VCO gain) was then calculated using:

$$K_{VCO} = \frac{f_{max} - f_{nominal}}{V_{max} - V_{min}}, \tag{26}$$

where f_{max} and $f_{nominal}$ are the maximum and nominal frequencies, and V_{max} and V_{min} are the corresponding input voltage limits. Figure 47 illustrates the output wave of the configured VCO at $V_{in} = 1$ corresponding to a frequency of 12 MHz.

TABLE XV
VCO PARAMETERS ADJUSTED ACCORDING TO THE APPLICATION REQUIREMENTS

Parameter	Value
KVCo (MHz/V)	2
Nominal frequency (MHz)	10
Minimum frequency (MHz)	10
Maximum frequency (MHz)	18

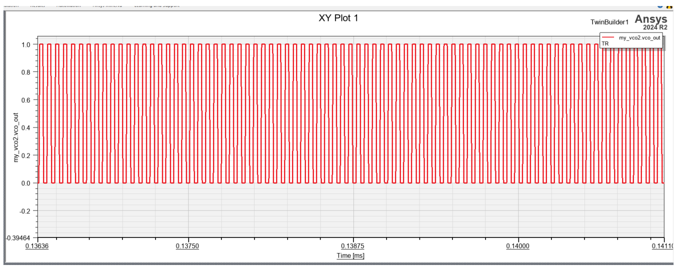

Fig. 47. The output Frequency of the VCO with Project specified Parameters

2) Main Logic and State Machine Implementation : The State Machine approach was utilized for designing the entire system. The first module, Frequency Measurement, employs two state machines: one for the edge counter and the other for the timer. These state machines bubble diagrams are illustrated in Figures 48 and 49, respectively. Synchronization between the Edge Counter and Timer is achieved through acknowledgment flags, `start_counter_ack_stable` and `counter_edge_ack_stable`, as discussed earlier.

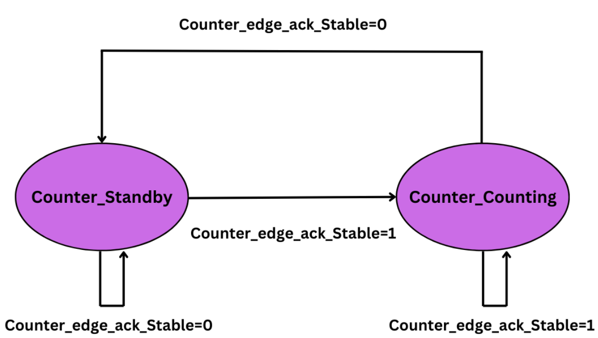

Fig. 48. State Machine Bubble Diagram for the Edge Counter

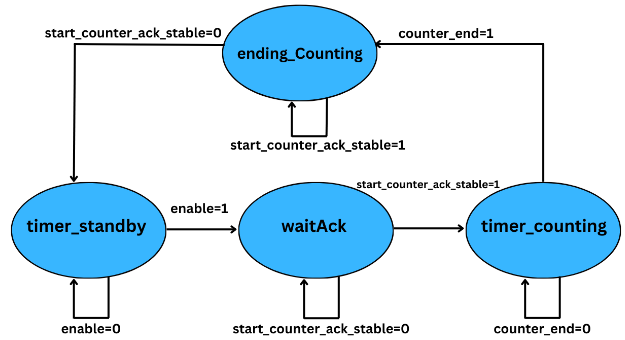

Fig. 49. State Machine Bubble Diagram for the Timer

The state machine bubble diagram for second module, the Acceleration Calculator (Decision Circuit), is shown in Figure 51

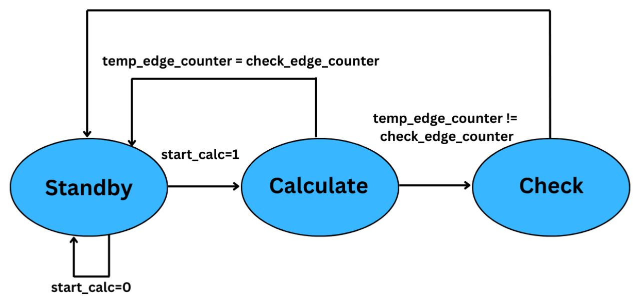

Fig. 50. State Machine Bubble Diagram for the Acceleration Calculator (Decision Circuit)

After the acceleration calculation is performed in the “calculate” state, the circuit compares the current and previous counted edges, saved in the “check” state. If the two counts are equal, the system transitions to the standby state, avoiding redundant recalculations, as the incoming signal frequency remains stable across two time windows. However, if the counts differ, the circuit recalculates and then transitions to the “check” state to save the current edge count for future comparisons.

The design was simulated using ModelSim software to validate its functionality and test the working principles of its components. Figure51 shows the state transitions in the three state machines (Timer, Counter, and Acceleration Calculator) in the design, where the synchronization of counters during resetting and starting operations can be observed. The signals `count_end` and `start_calc` are also depicted, fired at the end of each time window (when the timer reaches its limit) to stop counting and initiate the calculation process, respectively. Furthermore, the Figure highlights the final output of the design, which is the calculated acceleration. Figure52 provides a clearer, zoomed-in view of the waveform. Table XVI summarizes the measurement results for the two example frequencies shown in the figure.

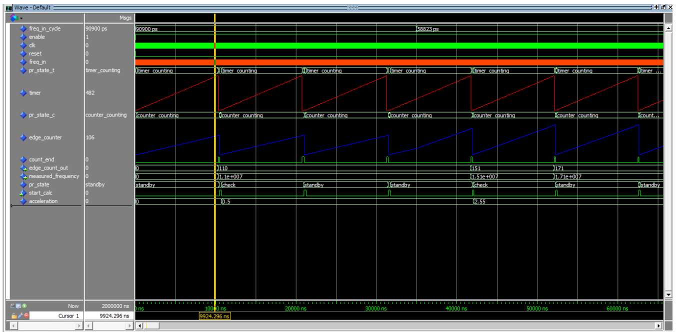

Fig. 51. Simulation Results showing values of edge counts, measured frequency, and acceleration

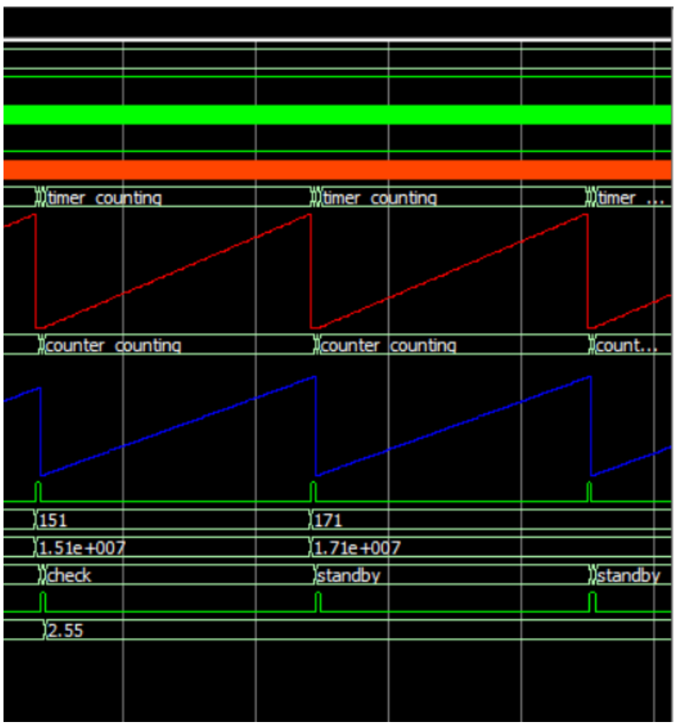

Fig. 52. Zoomed-in View of the Simulation Results Waveform.

TABLE XVI
COUNTED EDGES, MEASURED FREQUENCY, AND ACCELERATION

Counted Edges	Measured Frequency (MHz)	Acceleration (g)
110	1.1e+7	0.5
151	1.51e+7	2.55

3) Resource Analysis: The design was checked for synthesizability using Quartus software, considering the Cyclone III FPGA family. A resource utilization flow summary, shown in Figure 53, provides an overview of the analysis. Table XVII summarizes the resource utilization for the frequency measurement circuit. It highlights that the design requires only a minimal percentage of the available logic elements, logic registers, and combinational functions of the Cyclone III FPGA. However, the design utilizes 37 pins, primarily due to the 32-bit-wide output bus, used for storing the integer value of the frequency, which slightly increases the pin usage. Given its low overall resource requirements, this design could be implemented on an FPGA with fewer resources than those available in the Cyclone III family, reducing costs while ensuring efficient hardware utilization.

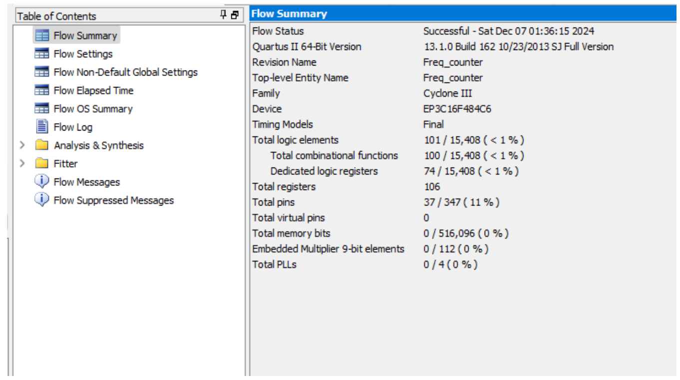

Fig. 53. Flow Summary from Quartus

TABLE XVII
RESOURCE UTILIZATION SUMMARY

Resource	Usage
Total combinational functions	100 / 15,408 (< 1%)
Dedicated logic registers	74 / 15,408 (< 1%)
Total registers	106
Total pins	37 / 347 (11%)
Total virtual pins, multiplier elements, PLLs	0

4) Synthesis and Register-Transfer Level (RTL): The Register-Transfer Level (RTL) view of the design is extracted from Quartus as shown in Figures 54. The diagram highlights the interaction between the two primary sub-modules: the Frequency Counter (MODULE1) and the Acceleration Calculator (MODULE2). Additionally, the RTL mapping, shown in Figure 60, 59 and 58 in the Annex, highlights how internal signals and logic blocks are connected. The layout includes combinational and sequential logic components such as multiplexers, arithmetic units, latches and flip-flops.

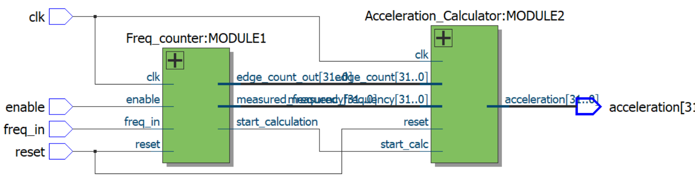

Fig. 54. RTL view of the top-level module as ectracted from Quartus

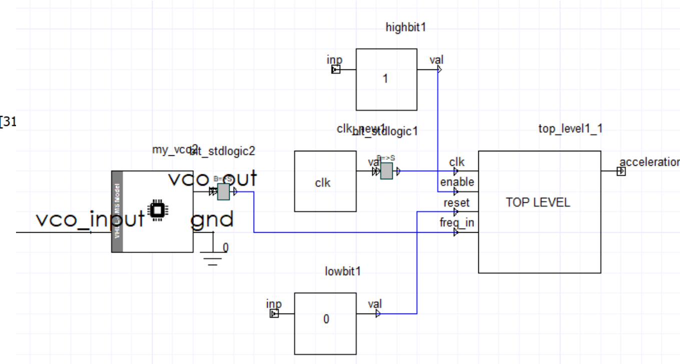

Fig. 55. Zoomed view of Digital Circuit in the Integrated System in Twinbuilder

5) Timing Analysis: Timing analysis was conducted to ensure there are no timing violations during operation. The analysis was performed through TimeQuest timing analysis with a Synopsys Design Constraints (SDC) file, where a global clock with a frequency of 50 MHz (20 ns cycle) and an input signal frequency of 17 MHz (58.82 ns cycle), which lies within the expected operating range. To simulate potential delays in the clock signal, another 50 MHz clock with a delay of 0.5 ns was introduced. Worst-case timing analysis is carried out to evaluate the robustness of a design, ensuring reliable performance even under the most adverse conditions, Figures61, 62 and 63 in the Annex. It helps identify potential challenges such as signal delays and clock skew, ensuring the system operates effectively in all scenarios. For our design, the hold and setup time were checked for the slow 1200mV 85C model and the summary of the analysis is shown in Table XVIII. The analysis confirmed a positive slack time of 12.28 ns for the input, ensuring that data remains stable before being required for computation. This result validates the timing robustness of the design under the given clock constraints.

The digital output of VCO was used as the `freq_in` signal for the digital component. A global clock signal with a frequency of 50 MHz, along with a high bit for the enable signal and a low bit for the reset signal, were designed as modules in VHDL-AMS and provided as inputs to the digital system, as illustrated in Figure 55. The results of the system-level integration are presented in Figure 56, showing the frequency output from the VCO plotted against the calculated acceleration value from the digital component. while Figure 57 represents the input acceleration for the system Vs the measured acceleration from the digital output, where the two values are approximately equal. These results confirm the correct functionality of the integrated system and validate the accuracy of the digital component in processing the VCO's output .

TABLE XVIII
SUMMARY OF SETUP AND HOLD TIMING ANALYSIS

Parameter	Main Clock	External Input
Setup slack (ns)	13.287	12.28
Hold slack (ns)	0.382	0.568

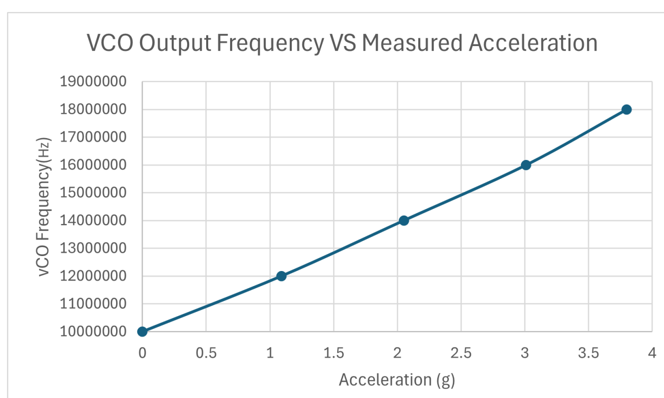

Fig. 56. System Level Result for VCO Frequency Output VS Digital Component Acceleration Output

6) Full System Integration and Validation: Following the validation of the digital design, full system integration was conducted using Twin Builder software to confirm the correctness of the overall digital circuit. The top-level digital component was integrated with the analog components.

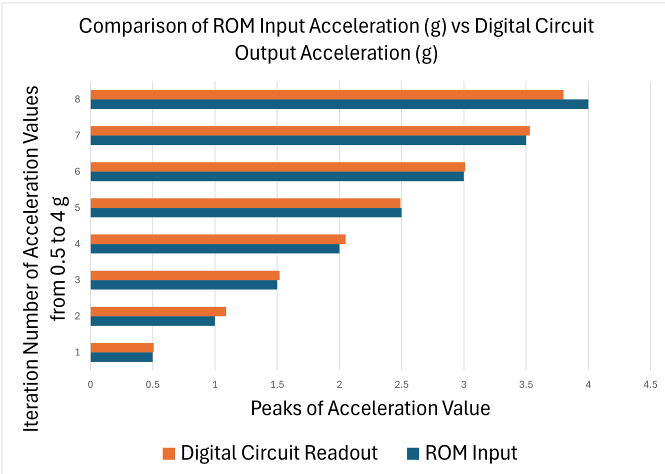

Fig. 57. System Level Result for Input Acceleration VS Digital Component Acceleration Output

VI. Conclusion of Overall Report

A Capacitive MEMS accelerometer was designed for range of motion monitoring application using Finite-element-modeling and model is further optimized using reduced-order-modeling (ROM).The output deflection and capacitance of the model is compared and analyzed with nodal displacements and matches well with desired characteristics. To realize the physical output, an analog circuitry with two-stage amplifier is designed to provide the sufficient gain and voltage sensitivity in terms of mV/G. The change in voltage as per input acceleration of the device is sensed by digital frequency measurement circuit and then converted back into acceleration (g) value based on decision circuit. The overall design and analysis of sensor presents the modeling, statistical and numerical approaches in engineering domain. The successful realization of output parameters such as gain, and bandwidth of 5 KHz ensures the measurement of patient activity during the rehabilitation process.

VII. Annex

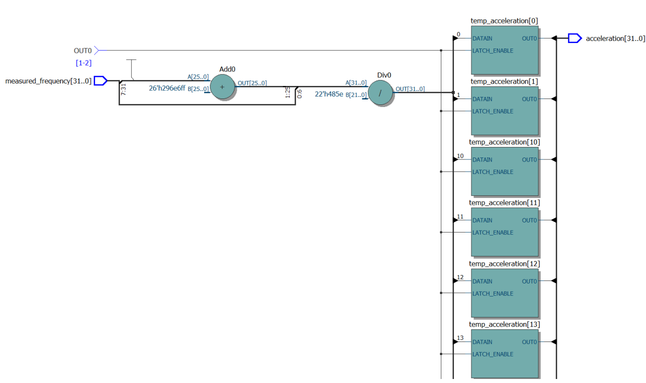

Fig. 58. RTL view of Digital Design with Arithmetic Operations in Acceleration Calculator

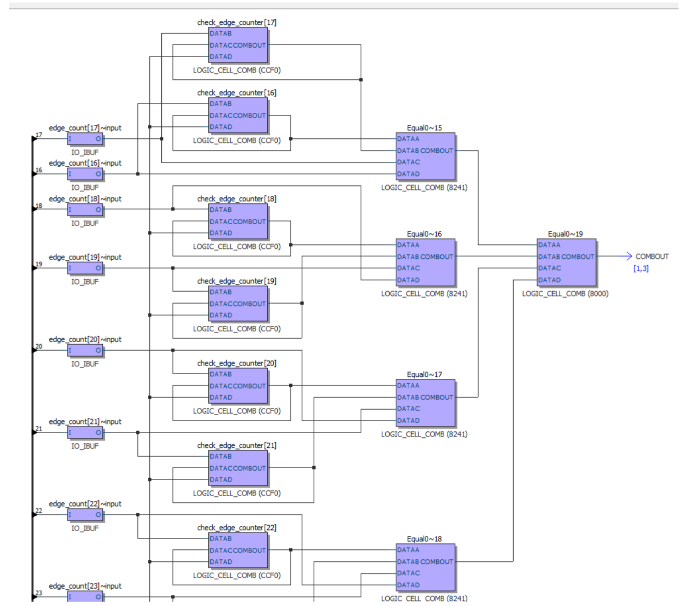

Fig. 59. RTL Zoomed View of Digital Design

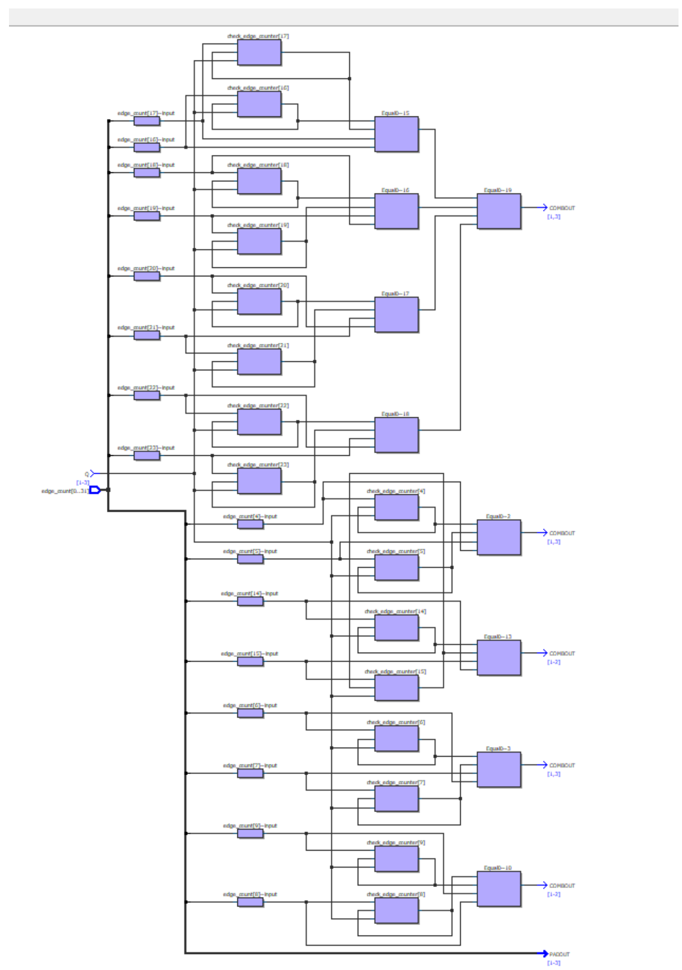

Fig. 60. RTL Full View of Digital Design

Slow 1200mV 85C Model Hold: 'clk'

	Slack	From Node	To Node	Launch Clock	Latch Clock	Relationship	Clock Skew	Data Delay
1	0.382	pr_state_t.timer_standby	pr_state_t.timer_standby	clk	clk	0.000	0.038	0.577
2	0.382	pr_state_t.ending_counting	pr_state_t.ending_counting	clk	clk	0.000	0.038	0.577
3	0.382	pr_state_t.timer_counting	pr_state_t.timer_counting	clk	clk	0.000	0.038	0.577
4	0.385	count_end	count_end	clk	clk	0.000	0.038	0.580
5	0.415	pr_state_t.timer_counting	pr_state_t.ending_counting	clk	clk	0.000	0.044	0.616
6	0.473	count_end	edge_count_out[9]~reg0	clk	clk	0.000	0.826	1.400
7	0.506	meta_clk_1	start_counter_ack_stable~_emulated	clk	clk	0.000	0.044	0.707
8	0.545	reset	count_end	clk_ext	clk	0.000	1.606	1.828
9	0.549	count_end	pr_state_t.ending_counting	clk	clk	0.000	0.090	0.796
10	0.566	count_end	pr_state_t.timer_counting	clk	clk	0.000	0.090	0.813
11	0.569	timer[29]	timer[29]	clk	clk	0.000	0.062	0.788
12	0.569	timer[19]	timer[19]	clk	clk	0.000	0.062	0.788
13	0.569	timer[15]	timer[15]	clk	clk	0.000	0.061	0.787
14	0.569	timer[13]	timer[13]	clk	clk	0.000	0.061	0.787
15	0.569	timer[3]	timer[3]	clk	clk	0.000	0.061	0.787
16	0.570	timer[31]	timer[31]	clk	clk	0.000	0.062	0.789
17	0.570	timer[27]	timer[27]	clk	clk	0.000	0.062	0.789
18	0.570	timer[21]	timer[21]	clk	clk	0.000	0.062	0.789
19	0.570	timer[17]	timer[17]	clk	clk	0.000	0.062	0.789
20	0.570	timer[11]	timer[11]	clk	clk	0.000	0.061	0.788
21	0.570	timer[1]	timer[1]	clk	clk	0.000	0.061	0.788
22	0.571	timer[22]	timer[22]	clk	clk	0.000	0.062	0.790
23	0.571	timer[16]	timer[16]	clk	clk	0.000	0.062	0.790
24	0.572	timer[25]	timer[25]	clk	clk	0.000	0.062	0.791
25	0.572	timer[23]	timer[23]	clk	clk	0.000	0.062	0.791
26	0.572	timer[18]	timer[18]	clk	clk	0.000	0.062	0.791
27	0.572	timer[9]	timer[9]	clk	clk	0.000	0.061	0.790
28	0.573	timer[30]	timer[30]	clk	clk	0.000	0.062	0.792
29	0.573	timer[14]	timer[14]	clk	clk	0.000	0.061	0.791
30	0.574	timer[28]	timer[28]	clk	clk	0.000	0.062	0.793
31	0.574	timer[26]	timer[26]	clk	clk	0.000	0.062	0.793
32	0.574	timer[24]	timer[24]	clk	clk	0.000	0.062	0.793
33	0.574	timer[20]	timer[20]	clk	clk	0.000	0.062	0.793
34	0.574	timer[12]	timer[12]	clk	clk	0.000	0.061	0.792
35	0.574	timer[10]	timer[10]	clk	clk	0.000	0.061	0.792
36	0.628	pr_state_t.timer_counting	count_end	clk	clk	0.000	0.008	0.793
37	0.630	pr_state_t.ending_counting	pr_state_t.timer_standby	clk	clk	0.000	0.044	0.831
38	0.726	pr_state_t.timer_counting	edge_count_out[9]~reg0	clk	clk	0.000	0.786	1.613
39	0.762	count_end	edge_count_out[3]~reg0	clk	clk	0.000	0.818	1.681
40	0.762	count_end	edge_count_out[11]~reg0	clk	clk	0.000	0.818	1.681
41	0.762	count_end	edge_count_out[14]~reg0	clk	clk	0.000	0.818	1.681
42	0.762	count_end	edge_count_out[21]~reg0	clk	clk	0.000	0.818	1.681
43	0.823	reset	pr_state_t.timer_standby	clk_ext	clk	0.000	1.648	2.148
44	0.827	reset	pr_state_t.timer_counting	clk_ext	clk	0.000	1.648	2.152

Fig. 61. Worst-case Timing Analysis Result for Hold Time of Clock

Slow 1200mV 85C Model Hold: 'freq_in'

	Slack	From Node	To Node	Launch Clock	Latch Clock	Relationship	Clock Skew	Data Delay
1	0.568	edge_counter[15]	edge_counter[15]	freq_in	freq_in	0.000	0.062	0.787
2	0.568	edge_counter[13]	edge_counter[13]	freq_in	freq_in	0.000	0.062	0.787
3	0.568	edge_counter[3]	edge_counter[3]	freq_in	freq_in	0.000	0.062	0.787
4	0.569	edge_counter[29]	edge_counter[29]	freq_in	freq_in	0.000	0.062	0.788
5	0.569	edge_counter[19]	edge_counter[19]	freq_in	freq_in	0.000	0.062	0.788
6	0.569	edge_counter[11]	edge_counter[11]	freq_in	freq_in	0.000	0.062	0.788
7	0.569	edge_counter[5]	edge_counter[5]	freq_in	freq_in	0.000	0.062	0.788
8	0.569	edge_counter[1]	edge_counter[1]	freq_in	freq_in	0.000	0.062	0.788
9	0.570	edge_counter[31]	edge_counter[31]	freq_in	freq_in	0.000	0.062	0.789
10	0.570	edge_counter[27]	edge_counter[27]	freq_in	freq_in	0.000	0.062	0.789
11	0.570	edge_counter[21]	edge_counter[21]	freq_in	freq_in	0.000	0.062	0.789
12	0.570	edge_counter[17]	edge_counter[17]	freq_in	freq_in	0.000	0.062	0.789
13	0.570	edge_counter[6]	edge_counter[6]	freq_in	freq_in	0.000	0.062	0.789
14	0.571	edge_counter[22]	edge_counter[22]	freq_in	freq_in	0.000	0.062	0.790
15	0.571	edge_counter[16]	edge_counter[16]	freq_in	freq_in	0.000	0.062	0.790
16	0.571	edge_counter[9]	edge_counter[9]	freq_in	freq_in	0.000	0.062	0.790
17	0.571	edge_counter[7]	edge_counter[7]	freq_in	freq_in	0.000	0.062	0.790
18	0.572	edge_counter[25]	edge_counter[25]	freq_in	freq_in	0.000	0.062	0.791
19	0.572	edge_counter[23]	edge_counter[23]	freq_in	freq_in	0.000	0.062	0.791
20	0.572	edge_counter[18]	edge_counter[18]	freq_in	freq_in	0.000	0.062	0.791
21	0.572	edge_counter[14]	edge_counter[14]	freq_in	freq_in	0.000	0.062	0.791
22	0.572	edge_counter[2]	edge_counter[2]	freq_in	freq_in	0.000	0.062	0.791
23	0.573	edge_counter[30]	edge_counter[30]	freq_in	freq_in	0.000	0.062	0.792
24	0.573	edge_counter[12]	edge_counter[12]	freq_in	freq_in	0.000	0.062	0.792
25	0.573	edge_counter[10]	edge_counter[10]	freq_in	freq_in	0.000	0.062	0.792
26	0.573	edge_counter[8]	edge_counter[8]	freq_in	freq_in	0.000	0.062	0.792
27	0.573	edge_counter[4]	edge_counter[4]	freq_in	freq_in	0.000	0.062	0.792
28	0.574	edge_counter[28]	edge_counter[28]	freq_in	freq_in	0.000	0.062	0.793
29	0.574	edge_counter[26]	edge_counter[26]	freq_in	freq_in	0.000	0.062	0.793
30	0.574	edge_counter[24]	edge_counter[24]	freq_in	freq_in	0.000	0.062	0.793
31	0.574	edge_counter[20]	edge_counter[20]	freq_in	freq_in	0.000	0.062	0.793
32	0.591	edge_counter[0]	edge_counter[0]	freq_in	freq_in	0.000	0.062	0.810
33	0.804	meta_freq_1	counter_edge_ack_stable~_emulated	freq_in	freq_in	0.000	0.062	1.023
34	0.841	edge_counter[15]	edge_counter[16]	freq_in	freq_in	0.000	0.064	1.062
35	0.843	edge_counter[13]	edge_counter[14]	freq_in	freq_in	0.000	0.062	1.062
36	0.843	edge_counter[1]	edge_counter[2]	freq_in	freq_in	0.000	0.062	1.062
37	0.843	edge_counter[3]	edge_counter[4]	freq_in	freq_in	0.000	0.062	1.062
38	0.844	edge_counter[5]	edge_counter[6]	freq_in	freq_in	0.000	0.062	1.063
39	0.844	edge_counter[17]	edge_counter[18]	freq_in	freq_in	0.000	0.062	1.063
40	0.844	edge_counter[29]	edge_counter[30]	freq_in	freq_in	0.000	0.062	1.063
41	0.844	edge_counter[11]	edge_counter[12]	freq_in	freq_in	0.000	0.062	1.063
42	0.844	edge_counter[19]	edge_counter[20]	freq_in	freq_in	0.000	0.062	1.063
43	0.845	edge_counter[21]	edge_counter[22]	freq_in	freq_in	0.000	0.062	1.064
44	0.845	edge_counter[9]	edge_counter[10]	freq_in	freq_in	0.000	0.062	1.064

Fig. 62. Worst-case Timing Analysis Result for Hold time of Input Frequency

Slow 1200mV 85C Model Setup: 'clk'

	Slack	From Node	To Node	Launch Clock	Latch Clock	Relationship	Clock Skew	Data Dela
1	13.287	timer[29]	edge_count_out[8]~reg0	clk	clk	20.000	-0.072	6.541
2	13.287	timer[29]	edge_count_out[16]~reg0	clk	clk	20.000	-0.072	6.541
3	13.287	timer[29]	edge_count_out[24]~reg0	clk	clk	20.000	-0.072	6.541
4	13.287	timer[29]	edge_count_out[30]~reg0	clk	clk	20.000	-0.072	6.541
5	13.287	timer[29]	edge_count_out[31]~reg0	clk	clk	20.000	-0.072	6.541
6	13.320	timer[19]	edge_count_out[8]~reg0	clk	clk	20.000	-0.072	6.508
7	13.320	timer[19]	edge_count_out[16]~reg0	clk	clk	20.000	-0.072	6.508
8	13.320	timer[19]	edge_count_out[24]~reg0	clk	clk	20.000	-0.072	6.508
9	13.320	timer[19]	edge_count_out[30]~reg0	clk	clk	20.000	-0.072	6.508
10	13.320	timer[19]	edge_count_out[31]~reg0	clk	clk	20.000	-0.072	6.508
11	13.388	timer[27]	edge_count_out[8]~reg0	clk	clk	20.000	-0.072	6.440
12	13.388	timer[27]	edge_count_out[16]~reg0	clk	clk	20.000	-0.072	6.440
13	13.388	timer[27]	edge_count_out[24]~reg0	clk	clk	20.000	-0.072	6.440
14	13.388	timer[27]	edge_count_out[30]~reg0	clk	clk	20.000	-0.072	6.440
15	13.388	timer[27]	edge_count_out[31]~reg0	clk	clk	20.000	-0.072	6.440
16	13.484	timer[29]	edge_count_out[0]~reg0	clk	clk	20.000	-0.072	6.344
17	13.484	timer[29]	edge_count_out[19]~reg0	clk	clk	20.000	-0.072	6.344
18	13.484	timer[29]	edge_count_out[20]~reg0	clk	clk	20.000	-0.072	6.344
19	13.484	timer[29]	edge_count_out[28]~reg0	clk	clk	20.000	-0.072	6.344
20	13.484	timer[29]	edge_count_out[29]~reg0	clk	clk	20.000	-0.072	6.344
21	13.498	timer[29]	edge_count_out[6]~reg0	clk	clk	20.000	-0.073	6.329
22	13.498	timer[29]	edge_count_out[17]~reg0	clk	clk	20.000	-0.073	6.329
23	13.498	timer[29]	edge_count_out[26]~reg0	clk	clk	20.000	-0.073	6.329
24	13.500	timer[21]	edge_count_out[8]~reg0	clk	clk	20.000	-0.072	6.328
25	13.500	timer[21]	edge_count_out[16]~reg0	clk	clk	20.000	-0.072	6.328
26	13.500	timer[21]	edge_count_out[24]~reg0	clk	clk	20.000	-0.072	6.328
27	13.500	timer[21]	edge_count_out[30]~reg0	clk	clk	20.000	-0.072	6.328
28	13.500	timer[21]	edge_count_out[31]~reg0	clk	clk	20.000	-0.072	6.328
29	13.517	timer[19]	edge_count_out[0]~reg0	clk	clk	20.000	-0.072	6.311
30	13.517	timer[19]	edge_count_out[19]~reg0	clk	clk	20.000	-0.072	6.311
31	13.517	timer[19]	edge_count_out[20]~reg0	clk	clk	20.000	-0.072	6.311
32	13.517	timer[19]	edge_count_out[28]~reg0	clk	clk	20.000	-0.072	6.311
33	13.517	timer[19]	edge_count_out[29]~reg0	clk	clk	20.000	-0.072	6.311
34	13.519	timer[18]	edge_count_out[8]~reg0	clk	clk	20.000	-0.072	6.309
35	13.519	timer[18]	edge_count_out[16]~reg0	clk	clk	20.000	-0.072	6.309
36	13.519	timer[18]	edge_count_out[24]~reg0	clk	clk	20.000	-0.072	6.309
37	13.519	timer[18]	edge_count_out[30]~reg0	clk	clk	20.000	-0.072	6.309
38	13.519	timer[18]	edge_count_out[31]~reg0	clk	clk	20.000	-0.072	6.309
39	13.531	timer[19]	edge_count_out[6]~reg0	clk	clk	20.000	-0.073	6.296
40	13.531	timer[19]	edge_count_out[17]~reg0	clk	clk	20.000	-0.073	6.296
41	13.531	timer[19]	edge_count_out[26]~reg0	clk	clk	20.000	-0.073	6.296
42	13.564	timer[30]	edge_count_out[8]~reg0	clk	clk	20.000	-0.072	6.264
43	13.564	timer[30]	edge_count_out[16]~reg0	clk	clk	20.000	-0.072	6.264

Fig. 63. Worst-case Timing Analysis Result for Setup time of Clock

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